



UNIVERSITÀ DEGLI STUDI DI PISA

Dipartimento di Matematica

Master's Degree in Mathematics

On Quantitative Stability for the Isoperimetric Inequality

Supervisors

Prof. Alessio Figalli
ETH Zürich

Prof. Luigi Ambrosio
Scuola Normale Superiore

Candidate

Matteo Gori

ACADEMIC YEAR 2025/2026

Abstract

This thesis provides an overview of known results concerning the quantitative Euclidean isoperimetric inequality and related functional inequalities. The primary focus is the investigation of the asymptotic behavior of the optimal constant as the dimension n tends to infinity.

We present an ongoing attempt to determine the precise asymptotic growth of this constant, following a roadmap proposed by Prof. Figalli:

- **Local Expansion:** Establishing the bound for sets satisfying weak graphicality conditions over the ball.
- **Local to Global:** Implementing a reduction procedure to substitute generic sets with the aforementioned graphical ones.

This framework is inspired by the approach successfully applied to the quantitative Sobolev inequality for $p = 2$ by Dolbeault, Esteban, Figalli, Frank and Loss, which is analyzed here in detail.

Acknowledgements

I would like to warmly thank my advisor Prof. Alessio Figalli for all the precious suggestions and insights he gave me during this last year. I am also extremely grateful to Prof. Luigi Ambrosio and Prof. Gian Maria Dall'Ara, who have been central figures in my academic path. I owe them most of my current mathematical interests. Finally, I would like to thank my friends Alberto Gervani, Alessandro Fenu, Leonardo Franchi, Laura Acinapura and Margherita Zucchelli for their invaluable academic and human support during the preparation of this thesis and especially throughout my excruciating PhD application period.

Contents

1	Introduction	4
1.1	History and State of the Art	4
1.2	Examples	6
2	The Known Approaches	9
2.1	Nearly-Spherical Case	9
2.2	Symmetrization Approach	12
2.3	Optimal Transport Method	18
2.4	Selection Principle	23
3	Further Isoperimetry	26
3.1	Finer Estimates	26
3.2	The Gaussian Case	28
3.3	On Manifolds	30
4	The Sobolev Case	32
4.1	Reductions	33
4.2	Local Case	34
4.2.1	First term	36
4.2.2	Second term	38
4.3	Local to Global	39
4.4	General Sign	41
5	The Current Attempt	42
5.1	Refining Fuglede	42
5.1.1	With Lipschitz Bound	43
5.1.2	With Volume Bound Only	45
5.2	From Local to Global	45
A	(Continuous) Symmetrizations	47
B	Finite Perimeter Sets	50

Chapter 1

Introduction

1.1 History and State of the Art

The sharp isoperimetric inequality states that, given a fixed volume $V > 0$ and a dimension n , the minimum value of the perimeter $P(E)$ among the finite perimeter sets $E \subset \mathbb{R}^n$ with volume $|E| = V$ is achieved only by the round balls. This result has been known in full generality since the work of De Giorgi, who proved it in 1958 (see [22]).

A question that has recently gained interest is whether a set with almost optimal perimeter, say $P(B) + \varepsilon$, is quantitatively close to a ball. The type of inequality on which we are going to focus is known as the Euclidean quantitative isoperimetric inequality, and states the following: for a set of finite perimeter $E \subset \mathbb{R}^n$ with $|E| < \infty$ and reference ball B with $|B| = |E|$, define the quantities

$$\delta(E) = \frac{P(E) - P(B)}{P(B)} \quad \alpha(E) = \inf \left\{ \frac{|E \Delta B(x, r)|}{|E|} : |E| = |B(x, r)| \right\}$$

where in particular $P(B) = n\omega_n^{\frac{1}{n}}|E|^{\frac{n-1}{n}}$. Then there exists a constant $C(n)$ such that

$$\alpha(E) \leq C(n)\sqrt{\delta(E)}$$

for all such E .

The optimality of the exponent has been achieved in full generality for the first time by Fusco, Maggi and Pratelli in 2008 (see [31]). However their proof involved many rough symmetrization arguments, with exponential constants arising, so that the resulting control on the optimal constant $C_{best}(n)$ was extremely weak. To be precise, their proof does not yield an explicit constant, due to the reliance on compactness arguments.

Before that, the best known exponent was $\frac{1}{4}$, achieved by Hall in [32], where he was also able to obtain the exponent $\frac{1}{2}$ for axially symmetric sets.

It is worth mentioning that, for $n = 2$, we are very close to determining the optimal minimizing shape for $\mathcal{R}(E) = \frac{\sqrt{\delta(E)}}{\alpha(E)}$, especially thanks to the works of Hall, Hayman, Campi, Cicalese and Leonardi (see [33], [14], [18], [19]).

The first explicit estimate on $C_{best}(n)$ was found by Figalli, Maggi and Pratelli in 2010 (see [27]), by expanding on Gromov's technique from the 1990s, which utilizes the Brenier map. By a sharp quantitative analysis of the map itself, they were able to obtain a growth control of $C_{best}(n) \lesssim n^{\frac{9}{2}}$ (actually improvable to n^3 as we will show).

However, the claimed growth rate is actually a bit smaller, specifically we expect

$$C_{best}(n) \asymp \sqrt{n}.$$

Recently Dolbeault, Esteban, Figalli, Frank and Loss determined the sharp asymptotic growth for the optimal constant of the quantitative 2-Sobolev inequality (see [23]). A very fruitful observation is that, up to a stereographic projection, the sharp 2-Sobolev inequality can be stated on the sphere $\mathbb{S}^n$ in such a way that one of the Aubin-Talenti bubbles becomes precisely the constant 1. In this setting, they divided the proof in two steps:

- **Local Expansion:** They proved the claimed quantitative inequality for functions $1 + r$ with $\|r\|_{L^{2^*}} \leq \delta$ for a fixed δ independent of the dimension n . The core idea is to truncate the function at specific heights to split the $\|1 + r\|_{L^{2^*}}$ term. This ensures that the small variations of r yield a quadratic contribution, which can then be controlled using spectral theory.
- **Local to Global:** It is well known that alternating radial rearrangements and a certain conformal transformation makes every function converge to an Aubin-Talenti bubble. Moreover one can partition the radial rearrangement into a sequence of suitable continuous Steiner symmetrizations, a tool first introduced by Brock in [12]. A careful study of this process makes us land back in the first case.

In the isoperimetric case, the first step might actually be carried out within a sufficiently large class of graphs over the ball. This was realized for the first time by Fuglede in [28], but he needed a very strict control of the C^1 norm. Borrowing the technique from Dolbeault et al. we aim to significantly weaken these hypotheses, but it is not clear yet if this is going to be enough to perform a local to global procedure.

The local to global step in the isoperimetric case would probably require a procedure reminiscent of that performed by Cicalese and Leonardi in [18]. A major limitation of their approach is that the proof proceeds by contradiction; thus, it only establishes the existence of such a constant.

1.2 Examples

Remark. By lemma 2.2.4 we can often assume the ball to be centered in the origin. Moreover, one can easily show that

$$\min_{|E|=|B(x,r)|} |E\Delta B| \leq 2 \min_{x,r} |E\Delta B|.$$

Hence we can consider any ball to bound $\alpha(E)$ from above, as long as we normalize with $|E|$ (or even with $|B|$ if $|E| \asymp |B|$).

The main examples that achieve the claimed growth rate can be split in two classes:

- Good deformations of the ball: precise computations for explicit examples are usually difficult to perform, but sometimes it is possible to make sharp estimates.

Let us consider an ellipsoid

$$E_\varepsilon = \left\{ \frac{x_1^2}{(1+\varepsilon)^2} + (1+\varepsilon)^{\frac{2}{n-1}} \sum_{i=2}^n x_i^2 \leq 1 : x \in \mathbb{R}^n \right\}$$

for which clearly $|E_\varepsilon| = \omega_n$. Borrowing the notation from the next chapter, we have

$$u(x) = \left(\frac{x_1^2}{(1+\varepsilon)^2} + (1+\varepsilon)^{\frac{2}{n-1}} \sum_{i=2}^n x_i^2 \right)^{-\frac{1}{2}} - 1 = \varepsilon v_n(x) + O_n(\varepsilon^2)$$

where $v_n(x) = \frac{n}{n-1} \left(x_1^2 - \frac{1}{n} \right)$. Now and when needed, we consider ε small enough to suppress the dependence on n , so that for example $O_n(\varepsilon^2) = o(\varepsilon)$. Then, with a computation analogous to Fuglede's estimate (see next chapter) we get

$$\begin{aligned} P(E_\varepsilon) - P(B) &= \frac{\varepsilon^2}{2} \left(\int_{\mathbb{S}^{n-1}} |\nabla v_n|^2 d\sigma - (n-1) \int_{\mathbb{S}^{n-1}} v_n^2 d\sigma \right) + o(\varepsilon^2) \\ &= \frac{\varepsilon^2}{2} (n+1) \int_{\mathbb{S}^{n-1}} v_n^2 d\sigma + o(\varepsilon^2) \end{aligned}$$

where we used that $\|\nabla v_n\|_2^2 = 2n \|v_n\|_2^2$, since v_n is a spherical harmonic of degree 2.

Analogously

$$\omega_n \alpha(E_\varepsilon) = \frac{1}{n} \int_{\mathbb{S}^{n-1}} |(1+u)^n - 1| d\sigma = \varepsilon \int_{\mathbb{S}^{n-1}} |v_n| d\sigma + o(\varepsilon)$$

It is easy to compute

$$\int_{\mathbb{S}^{n-1}} v_n^2 d\sigma = \frac{2n}{(n-1)(n+2)} \omega_n \sim \frac{2\omega_n}{n}$$

while for the other integral, we recall that the random variable $\sqrt{n}x_1$ (with normalized σ) converges in law to $\mathcal{N}(0, 1)$, so

$$\frac{1}{\omega_n} \int_{\mathbb{S}^{n-1}} |v_n| d\sigma \sim \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} |t^2 - 1| e^{-\frac{t^2}{2}} dt = \frac{4}{\sqrt{2\pi e}}$$

All in all $\alpha(E_\varepsilon) \sim \frac{4\varepsilon}{\sqrt{2\pi e}}$ and $\delta(E_\varepsilon) \sim \frac{\varepsilon^2}{n}$, and consequently

$$\frac{\alpha(E_\varepsilon)}{\sqrt{\delta(E_\varepsilon)}} \sim 4\sqrt{\frac{n}{2\pi e}}$$

which proves both the upper bound $\frac{1}{2}$ for the exponent and the lower bound $C_{best}(n) \gtrsim \sqrt{n}$.

- Big asymmetry: let $E = B(x, 2^{-\frac{1}{n}}) \cup B(-x, 2^{-\frac{1}{n}})$ and x large enough, then $\alpha(E) = \frac{1}{2}$ while

$$\delta(E) = \frac{2n\omega_n 2^{-\frac{n-1}{n}} - n\omega_n}{n\omega_n} = 2^{\frac{1}{n}} - 1 \sim \frac{\log(2)}{n}.$$

Finally, let us show an interesting sub-optimal case, which is that of a sphere with (possibly many) tentacles:

Let $\varepsilon(n) > 0$ be as small as needed, then attach to a unitary ball m disjoint cylinders with base radius ε and height $h \asymp 1$. Since their volume and surface area satisfy $V \asymp \omega_{n-1} \varepsilon^{n-1} h$ and $A \asymp (n-1)\omega_{n-1} \varepsilon^{n-2} h$, we can estimate

$$P(E) - P(B) \asymp m(n-1)\omega_{n-1} \varepsilon^{n-2} h \quad |E \setminus B| \asymp m\omega_{n-1} \varepsilon^{n-1} h$$

so, recalling that $\frac{\omega_{n-1}}{\omega_n} \sim \sqrt{\frac{n}{2\pi}}$, we find

$$\frac{\alpha(E)^2}{\delta(E)} \asymp m\sqrt{n}\varepsilon^n$$

One could naively expect m to be taken proportional to $\frac{n\omega_n}{\omega_{n-1}\varepsilon^{n-1}} \asymp \frac{\sqrt{n}}{\varepsilon^{n-1}}$, which is the number of bases required to match the area of $\mathbb{S}^{n-1}$, but we face two great issues:

- We are missing an ε at the denominator, which is required because surely $\mathbb{S}^{n-1}$ is not uniformly flat in n in such a way that allows us to approximately glue these cylinders (i.e., the geodesic balls differ from the Euclidean ones already for small radii).
- Even if ε could be taken to be arbitrarily small, we would still have issues. Indeed, the problem could be naturally formulated in $\mathbb{R}^{n-1}$, however the spherical packing constant is significantly smaller than the ratio between a set and a given ball.

Chapter 2

The Known Approaches

From now on, we will only consider finite perimeter sets with finite measure, and σ will be the (non-normalized) surface measure on $\mathbb{S}^{n-1}$.

Let us further remark that, since the asymmetry index satisfies the trivial bound $\alpha(E) \leq 2$ for every E , if we only consider sets such that $\delta(E) \geq \delta(n)$, a quantitative isoperimetric inequality holds in the form

$$\alpha(E) \leq \frac{2}{\sqrt{\delta(n)}} \sqrt{\delta(E)}.$$

Therefore if we prove an inequality $\alpha(E) \leq C(n) \sqrt{\delta(E)}$ for sets satisfying $\delta(E) \leq \delta(n)$, we actually get the inequality

$$\alpha(E) \leq \max \left\{ C(n), \frac{2}{\sqrt{\delta(n)}} \right\} \sqrt{\delta(E)}$$

for every set E .

For example, if we aim at proving our claimed bound $C_{best}(n) \lesssim \sqrt{n}$, we can freely restrict to sets satisfying $\delta(E) \lesssim \frac{1}{n}$, and any stronger condition would not be allowed.

2.1 Nearly-Spherical Case

One of the first strong results in this direction was obtained by Fuglede in 1989 (see [28]) for a suitable class of graphs over the ball. Specifically we will call nearly-spherical a set E that can be described as

$$\{tx(1 + u(x)) : t \in [0, 1], x \in \mathbb{S}^{n-1}\} \subset \mathbb{R}^n$$

for a function $u: \mathbb{S}^{n-1} \rightarrow (-1, +\infty)$ in $W^{1,\infty}(\mathbb{S}^{n-1})$ which satisfies the estimates $\|u\|_\infty \leq \frac{\varepsilon}{n}$ and $\|\nabla u\|_\infty \leq \varepsilon$ for a suitable $\varepsilon > 0$, $|E| = \omega_n$ and with barycenter in the origin, i.e.

$$\int_{\mathbb{S}^{n-1}} ((1 + u)^n - 1) \, d\sigma = 0 \quad \int_{\mathbb{S}^{n-1}} (1 + u)^{n+1} x \, d\sigma = 0.$$

Theorem 2.1.1 (Fuglede). *There exists a universal $\varepsilon > 0$ such that, for every n , every nearly spherical set satisfies*

$$P(E) - P(B) \geq \frac{1}{3\omega_n} |E\Delta B|^2.$$

Remark. The normalized version is $\alpha(E) \leq \frac{|E\Delta B|}{\omega_n} \leq \sqrt{3n}\sqrt{\delta(E)}$, which is coherent with our claim.

Throughout the proof, $C > 0$ denotes a universal constant (independent of n) whose exact value may change from line to line.

Proof. First, the area formula implies that

$$P(E) = \int_{\mathbb{S}^{n-1}} (1+u)^{n-1} \sqrt{1 + \frac{|\nabla u|^2}{(1+u)^2}} d\sigma$$

where ∇u is the tangential gradient. By the assumptions on u

$$\begin{aligned} (1+u)^{n-1} \sqrt{1 + \frac{|\nabla u|^2}{(1+u)^2}} &\geq (1+u)^{n-1} \left(1 + (1-C\varepsilon) \frac{1}{2} \frac{|\nabla u|^2}{(1+u)^2} \right) \\ &\geq (1+u)^{n-1} + (1-C\varepsilon) \frac{1}{2} |\nabla u|^2, \end{aligned}$$

and in addition

$$\begin{aligned} \int_{\mathbb{S}^{n-1}} ((1+u)^{n-1} - 1) d\sigma &= \int_{\mathbb{S}^{n-1}} ((1+u)^{n-1} - 1) - \frac{n-1}{n} \int_{\mathbb{S}^{n-1}} [(1+u)^n - 1] d\sigma \\ &\geq \left(-\frac{(n-1)^2}{2} + \frac{(n-1)(n-2)}{2} - C\varepsilon n \right) \int_{\mathbb{S}^{n-1}} u^2 d\sigma \\ &\geq -(1+C\varepsilon) \frac{n-1}{2} \int_{\mathbb{S}^{n-1}} u^2 d\sigma, \end{aligned}$$

where the estimate follows from the following observation: if we expand the powers and simplify, for degree ≥ 3 we get a sum of the form $\sum_{i=3}^n C_i(n)u^i$ where $C_i(n) \lesssim n^{i-1}$, therefore this value is bounded by

$$nu^2 \sum_{i=3}^n \varepsilon^{i-2} \leq \frac{\varepsilon}{1-\varepsilon} nu^2 \leq C\varepsilon nu^2$$

as we wanted.

Adding these two pieces yields

$$P(E) - P(B) \geq (1-C\varepsilon) \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\nabla u|^2 d\sigma - (1+C\varepsilon) \frac{n}{2} \int_{\mathbb{S}^{n-1}} u^2 d\sigma. \quad (2.1)$$

We are in need of an appropriate 2-Poincaré inequality, so as usual we rely on spherical harmonics tools (a standard reference is [20]). Let Y_i^j be the i -th degree

normalized spherical harmonics and $c_i^j = \langle Y_i^j, u \rangle$, then $-\Delta_{\mathbb{S}^{n-1}} Y_i^j = i(i+n-2)$, which implies that

$$\|\nabla u\|_{L^2(\sigma)}^2 = \sum_{i=1}^{\infty} i(i+n-2) \sum_{j=1}^{G(n,i)} (a_i^j)^2.$$

If we were able to prove that the parts of degree 0 and 1 are negligible, we would obtain a Poincaré-type inequality with coefficient close to $2n$: indeed, analogously to before, we have

$$0 = \int_{\mathbb{S}^{n-1}} [(1+u)^n - 1] d\sigma \leq n \int_{\mathbb{S}^{n-1}} u d\sigma + (1+C\varepsilon) \binom{n}{2} \int_{\mathbb{S}^{n-1}} u^2 d\sigma,$$

which yields

$$|a_0| = \left| \int_{\mathbb{S}^{n-1}} u d\sigma \right| \leq (1+C\varepsilon) \frac{n-1}{2} \int_{\mathbb{S}^{n-1}} u^2 \leq (1+C\varepsilon) \frac{n}{2} \frac{\varepsilon}{n} \|u\|_1 \leq \varepsilon \|u\|_2,$$

and analogously, given ℓ the (normalized) projection of u on degree 1 harmonics,

$$\begin{aligned} \sqrt{\sum_{j=1}^{G(n,1)} |a_1^j|^2} &= \left| \int_{\mathbb{S}^{n-1}} u \ell d\sigma \right| \leq \frac{n}{2} \left| \int_{\mathbb{S}^{n-1}} u^2 (1 + \varepsilon(u)) \ell d\sigma \right| \\ &\leq \frac{n}{2} \sqrt{\int_{\mathbb{S}^{n-1}} u^4 (1 + \varepsilon(u))^2 d\sigma} \leq (1+C\varepsilon) \frac{n}{2} \frac{\varepsilon}{n} \|u\|_2 \leq \varepsilon \|u\|_2. \end{aligned}$$

respectively thanks to the volume and barycenter condition (here $\varepsilon(u)$ denotes a function of u satisfying $|\varepsilon(u)| \leq C\varepsilon$). In turn, since $i(i+n-2) \geq 2n$ for $i \geq 2$,

$$\|u\|_{L^2}^2 \leq \frac{1}{2n(1-C\varepsilon)} \|\nabla u\|_{L^2}^2,$$

and if we plug it into 2.1 we get

$$P(E) - P(B) \geq (1-C\varepsilon) \frac{1}{4} \|\nabla u\|_2^2 \geq (1-C\varepsilon) \frac{n}{2} \|u\|_{L^2}^2 \geq (1-C\varepsilon) \frac{1}{2\omega_n} \|u\|_{L^1}^2,$$

which yields the desired conclusion after noticing that

$$|E\Delta B| = \int_{\mathbb{S}^{n-1}} \frac{|(1+u)^n - 1|}{n} d\sigma \leq (1+C\varepsilon) \int_{\mathbb{S}^{n-1}} |u| d\sigma.$$

□

As mentioned in the introduction, this theorem proves the claimed inequality for a small class of sets, but one might hope that they are a good model to understand the problem even in the general setting.

2.2 Symmetrization Approach

The first complete proof of the quantitative isoperimetric inequality has been carried out by Fusco, Maggi and Pratelli in [31], and it was further refined by Maggi in [37]. It relies on a reduction to sets enjoying progressively more symmetries:

1. Uniform boundedness;
2. n -symmetry: sets which are symmetric with respect to every coordinate hyperplane;
3. Axial symmetry: sets described as

$$\{(x', y) \in \mathbb{R}^{n-1} \times \mathbb{R} : |x'| \leq r(y)\} = \bigcup_{y \in \mathbb{R}} B(0, r(y)) \times \{y\}$$

for a suitable non-negative function r ; in particular they are invariant under Schwarz symmetrization.

We will not keep track of the constants because some of them are not found constructively, and the constructive ones are often not sharp.

Moreover, in all the statements in this section it is implicitly assumed that the thesis holds for sets satisfying an appropriate bound $\delta(E) \leq \delta(n)$.

Lemma 2.2.1. *There exists a constant $\ell(n) > 0$ such that for every set of finite perimeter E with $|E| = \omega_n$ there exists $F \subset Q_\ell = [-\ell, \ell]^n$ with $|F| = |E|$ such that*

$$\delta(F) \leq C(n)\delta(E) \quad \alpha(E) \leq \alpha(F) + C(n)\delta(E).$$

Proof. We will prove that E can be restricted in any fixed direction, so that the thesis will trivially follow from restricting with respect to an orthonormal basis.

Up to tilting E , we can assume that $\mathcal{H}^{n-1}(\{v_E = \pm e_n\}) = 0$ so that in particular $v_E(t) = \mathcal{H}^{n-1}(E \cap \{x_n = t\}) \in W^{1,1}(\mathbb{R})$. Let $g(t) = \frac{|E \cap \{x_n < t\}|}{\omega_n}$ which is in C^1 with $g'(t) = \frac{v_E(t)}{\omega_n}$. By the sharp isoperimetric inequality we get

$$P(E \cap \{x_n < t\}) \geq n\omega_n g(t)^{\frac{n-1}{n}} \quad P(E \cap \{x_n \geq t\}) \geq n\omega_n (1 - g(t))^{\frac{n-1}{n}},$$

which, together with observing that

$$P(E \cap \{x_n < t\}) \leq P(E, \{x_n < t\}) + v_E(t) \quad P(E \cap \{x_n \geq t\}) \leq P(E, \{x_n > t\}) + v_E(t).$$

yields

$$v(E) \geq \frac{1}{2}n\omega_n (g(t)^{\frac{n-1}{n}} + (1 - g(t))^{\frac{n-1}{n}} - 1 - \delta(E)). \quad (2.2)$$

Now define $\psi(t) = t^{\frac{n-1}{n}} + (1-t)^{\frac{n-1}{n}} - 1$ which by concavity satisfies $\psi(t) \geq 2(2^{\frac{1}{n}} - 1)t$ for $t \in [0, \frac{1}{2}]$. Then we can assume that $2\delta(E) < \psi(\frac{1}{2})$ (otherwise the thesis trivially

holds) and we can fix $t_1 < t_2$ such that $g(t_1) = 1 - g(t_2)$ and $\psi(g(t_1)) = \psi(g(t_2)) = 2\delta(E)$. Note that $\psi(g(t)) \geq 2\delta(E)$ for $t \in [t_1, t_2]$ by concavity, which plugged into 2.2 gives

$$v_E(t) \geq \frac{n\omega_n}{4}\psi(g(t)),$$

and consequently

$$t_2 - t_1 \leq \frac{4}{n} \int_{t_1}^{t_2} \frac{g'(t)}{\psi(g(t))} dt = \frac{4}{n} \int_{g(t_1)}^{g(t_2)} \frac{1}{\psi(s)} ds \leq \frac{4}{n} \int_0^1 \frac{1}{\psi(s)} ds = C(n).$$

Furthermore, let

$$\tau_1 = \max \left\{ t \leq t_1 : v_E(t) \leq \frac{n\omega_n\delta(E)}{2} \right\} \quad \tau_2 = \min \left\{ t \geq t_2 : v_E(t) \leq \frac{n\omega_n\delta(E)}{2} \right\},$$

which satisfy

$$t_1 - \tau_1 \leq \frac{2}{n\delta(E)} \int_{t_1}^{\tau_1} g'(t) dt \leq \frac{2g(t_1)}{n\delta(E)} \leq \frac{2}{n(2^{\frac{1}{n}} - 1)} \lesssim 1$$

thanks to $g(t_1) \leq \frac{\delta(E)}{2^{\frac{1}{n}} - 1}$ (and the same goes for τ_2). Hence the strip $E' = E \cap \{\tau_1 \leq x_1 \leq \tau_2\}$ satisfies

$$|E'| \geq |E| (1 - 2g(t_1)) \geq |E| \left(1 - 2\frac{\delta(E)}{2^{\frac{1}{n}} - 1} \right) \quad P(E') \leq P(E) + n\omega_n\delta(E).$$

Now let $F = rE'$ with $r = \left(\frac{|E|}{|E'|} \right)^{\frac{1}{n}}$ (so that $|F| = |E|$) and let us check that it has the desired properties: up to taking $\delta(E)$ small enough (in terms of n) we get

$$r \leq \left(1 - 2\frac{\delta(E)}{2^{\frac{1}{n}} - 1} \right)^{-\frac{1}{n}} \leq 1 + C(n)\delta(E),$$

hence

$$\begin{aligned} P(F) &= r^{n-1}P(E') \leq (1 + C(n)\delta(E))(P(E) + n\omega_n\delta(E)) \\ &\leq n\omega_n(1 + C(n)\delta(E))(1 + 2\delta(E)) \leq n\omega_n(1 + C(n)\delta(E)). \end{aligned}$$

Finally, if B is a minimizing ball for $\alpha(F)$ (assume it is centered in the origin), we get

$$\begin{aligned} n\omega_n\alpha(E) &\leq |E\Delta B| \leq |E\Delta E'| + \left| E'\Delta B_{\frac{1}{r}} \right| + \left| B_{\frac{1}{r}}\Delta B \right| \\ &= \frac{n\omega_n\alpha(F)}{r^n} + |E \setminus E'| + \left| B \setminus B_{\frac{1}{r}} \right| \leq n\omega_n\alpha(F) + C(n)\delta(E) \end{aligned}$$

□

Next, let us reduce to n -symmetric sets. The idea will be to iteratively gain new symmetries by reflecting our set E . To be precise, let $H = \{x \cdot \nu = 0\}$ be a hyperplane that cuts E in half, i.e. $|E^1| = |E^2| = \frac{|E|}{2}$ where $E^1 = E \cap \{x \cdot \nu \geq 0\}$ and $E^2 = E \cap \{x \cdot \nu \leq 0\}$. Moreover, let r be the reflection over H and define $E_s^i = E^i \cup r(E^i)$.

Then looking at the perimeters we have

$$P(E^1) + P(E^2) \leq 2P(E) \implies \delta(E^i) \leq 2\delta(E)$$

for both $i = 1, 2$, where the implication follows from the sharp isoperimetric inequality.

If it were true that $\alpha(E) \lesssim \alpha(E^i)$ for either $i = 1, 2$, we could just substitute E with E^i and gain one hyperplane of symmetry. However, this is completely false in general: for example, consider two half-balls with base on the same hyperplane.

Luckily this issue can be fixed by taking two orthogonal hyperplanes, as we now show. We first need a technical lemma.

Lemma 2.2.2. *Fix any $\ell > 0$, then for every $\varepsilon > 0$ there exists a $\delta > 0$ such that if $E \subset Q_\ell$ satisfies $\delta(E) \leq \delta$, then $\alpha(E) \leq \varepsilon$.*

Proof. By contradiction there exists an $\varepsilon > 0$ and a sequence $E_n \subset Q_\ell$ such that $\alpha(E_n) \geq \varepsilon$ and $\delta(E_n) \rightarrow 0$. Then, up to a subsequence, there exists a finite perimeter set $E \subset Q_\ell$ such that $E_n \rightarrow E$ in measure, hence satisfying $\alpha(E) \geq \varepsilon$. Moreover, by lower semi-continuity of the perimeter $\delta(E) = 0$, therefore E is a round ball by the sharp isoperimetric inequality, in contradiction with $\alpha(E) > 0$. $\square$

Lemma 2.2.3. *If $E \subset Q_\ell$, $|E| = \omega_n$ and H_1, H_2 are two orthogonal hyperplanes that cut E in four parts of equal measure, at least one of the sets E_i^j satisfies $\alpha(E) \lesssim \alpha(E_i^j)$.*

We will provide a slight modification of the original proof. These equivalences are justified by the following principle: if a value is proportional to $\alpha(E)$, modifying a defining term by $C(n)\alpha(E)$ maintains the equivalence for sufficiently small $C(n)$.

Proof Sketch. By a similar argument to lemma 2.2.6 below, we can assume that there exists a unitary ball B_i^j centered in $H_1 \cap H_2$ such that $\omega_n \alpha(E_i^j) \asymp |E_i^j \Delta B_i^j|$ for all i, j . Further assume that $|E_i^j \Delta B_i^j| \leq C(n)\alpha(E)$ for some constant $C(n)$ to be fixed. If $C(n)$ and $\alpha(E)$ are small enough, we have $\alpha(E) \asymp |B_1^1 \Delta B_1^2|$ and

$$C(n)\alpha(E) \geq |(E \Delta B_2^1) \cap H_2^1| \asymp |((B_1^1 \cap H_1^1) \cup (B_1^2 \cap H_1^2)) \Delta B_2^1| \cap H_2^1| \asymp \alpha(E),$$

where the last term is $\asymp \alpha(E)$ by the first observation. For $C(n)$ small enough this gives a contradiction. $\square$

To obtain the last symmetry we will need the following lemma on n -symmetric sets.

Lemma 2.2.4. *If E is n -symmetric, then $\frac{|E\Delta B|}{|E|} \leq 3\alpha(E)$, and if it is also convex then $\alpha(E) = \frac{|E\Delta B|}{|E|}$.*

Proof. Let B_x be a minimizing ball for $\alpha(E)$, then

$$\begin{aligned} |E\Delta B| &\leq |E\Delta B_x| + |B_x\Delta B| \leq |E\Delta B_x| + |B_x\Delta B_{-x}| \\ &\leq |E\Delta B_x| + |E\Delta B_x| + |E\Delta B_{-x}| = 3|E\Delta B_x|. \end{aligned}$$

Note that we only required symmetry with respect to the origin. The second statement easily follows from $|E\cap F| \leq |E^S\cap F^S|$ where E^S is the Steiner symmetral of E . $\square$

Lemma 2.2.5. *Given an orthonormal basis $\{e_1, \dots, e_n\}$ such that E is $(n-1)$ -symmetric with respect to $\{e_1, \dots, e_{n-1}\}$ and $\{x_n = 0\}$ cut E in half, then at least one among E_n^1 and E_n^2 satisfies $\alpha(E) \lesssim \alpha(E_n^i)$.*

Proof. The previous lemma implies that

$$\begin{aligned} |E\Delta B| &= |(E\Delta B) \cap \{x_n \geq 0\}| + |(E\Delta B) \cap \{x_n \leq 0\}| \\ &= \frac{1}{2}(|E_n^1\Delta B| + |E_n^2\Delta B|) \leq \frac{3}{2}(\alpha(E_n^1) + \alpha(E_n^2)), \end{aligned}$$

from which the thesis follows. $\square$

Remark. An alternative approach for this step was investigated, but it turned out to be flawed. The idea was the following:

Up to a translation, assume that E is cut in half by all the coordinate hyperplanes. Call E_j the 2^n different intersections of E with the quadrants C_j and E_j^s the n -reflections of these sets, for example

$$\chi_{E_1^s}(x_1, \dots, x_n) = \chi_{E_1}(|x_1|, \dots, |x_n|)$$

if C_1 is the positive quadrant. Then, as before, we have

$$\sum_{j=1}^{2^n} P(E_j^s) \leq 2^n P(E),$$

but this time this inequality does not imply that $\delta(E_j) \leq 2^n \delta(E)$. In fact, there may not be a choice of hyperplanes such that the E_i all have the same measure $2^{-n}|E|$ (this is known as Grünbaum partition problem). So if we estimate the $P(E_i)$'s with the sharp isoperimetric inequality, we get a sum of the form $\sum_{i=1}^{2^n} f(|E_i|)$ where $f(t) = t^{\frac{n-1}{n}}$ is concave, hence it is maximal (as we would need) only when the $|E_i|$'s are all equal.

Moreover we also face issues when dealing with asymmetries. One could hope to use the following lemma to relate E^i to E .

Lemma 2.2.6. *There exists $\varepsilon(n) > 0$ such that, if a set E is cut in half by every coordinate hyperplane and $\alpha(E) \leq \varepsilon$, then $\frac{|E \Delta B|}{|E|} \lesssim \alpha(E)$.*

Proof. Up to rescaling assume that $|E| = \omega_n$. Let $H = \{x_i = 0\}$, then

$$|B(x) \cap H^+| - |(E \Delta B(x)) \cap H^+| \leq |E \cap H^+| = |E \cap H^-| \leq |B(x) \cap H^-| + |(E \Delta B(x)) \cap H^-|,$$

which together with the reversed inequality yields

$$\left| |B(x) \cap H^+| - \frac{\omega_n}{2} \right| \leq \omega_n \alpha(E).$$

Hence, for $\varepsilon(n)$ small enough, we get $|x_i| \lesssim \alpha(E)$ for all i , so that

$$|x| \lesssim \alpha(E) \implies |B(x) \Delta B| \lesssim \alpha(E)$$

and finally $|E \Delta B| \leq |E| \alpha(E) + |B \Delta B(x)| \lesssim |E| \alpha(E)$. $\square$

Since we have $|E_i^s \Delta B| \leq 2^n |E \Delta B| \lesssim \alpha(E)$ the E_i^s are somewhat round, but their measure might differ from ω_n by an $\alpha(E)$ fraction, therefore $|E_i^s \Delta B|$ is not necessarily proportional to $\omega_n \alpha(E_i^s)$ (and it is easy to build counterexamples consisting of spherical wedges).

At this point we restricted ourselves to n -symmetric sets $E \subset Q_{2\ell}$. Further restricting to axially symmetric sets presents some technical difficulties, as we now show.

Lemma 2.2.7. *If a quantitative isoperimetric inequality holds in $\mathbb{R}^{n-1}$, for every n -symmetric set $E \subset Q_\ell$ with $|E| = \omega_n$, its Schwarz symmetral E^* satisfies*

$$|E \Delta E^*| \leq C(n) \sqrt{\delta(E)}.$$

Proof. The proof goes by induction, starting from $n = 1$ where it is trivially true. In dimension n let $E_t = E \cap \{x_1 = t\}$ and (by approximation) assume that $\mathcal{H}^{n-1}(\{v_E^1 = \pm 1\}) = 0$, then we have

$$\begin{aligned} P(E) - P(B) &\geq P(E) - P(E^*) \\ &\geq \int_{\mathbb{R}} \sqrt{P(E_t)^2 + v_E'(t)^2} - \sqrt{P(E_t^*)^2 + v_E'(t)^2} dt \\ &\geq \int_{\mathbb{R}} \frac{P(E_t)^2 - P(E_t^*)^2}{\sqrt{P(E_t)^2 + v_E'(t)^2} + \sqrt{P(E_t^*)^2 + v_E'(t)^2}} dt \\ &\geq \left(\int_{\mathbb{R}} \sqrt{P(E_t)^2 - P(E_t^*)^2} dt \right)^2 \frac{1}{P(E) + P(E^*)}, \end{aligned}$$

and in addition $\delta(E) \leq 1 \implies P(E^*) \leq P(E) \leq 2P(B)$, hence

$$\begin{aligned} \sqrt{\delta(E)} &\gtrsim \int \sqrt{P(E_t)^2 - P(E_t^*)^2} dt \gtrsim \int P(E_t^*) \sqrt{\delta(E_t)} \\ &\gtrsim \int \frac{P(E_t^*)}{\mathcal{H}^{n-1}(E_t^*)} \mathcal{H}^{n-1}(E_t^* \Delta E_t) dt \gtrsim \int \mathcal{H}^{n-1}(E_t \Delta E_t^*) = |E_t \Delta E_t^*|. \end{aligned}$$

where the second step follows from the quantitative isoperimetric inequality in $\mathbb{R}^{n-1}$, the third one from 2.2.4 and the fourth one from the uniform boundedness. $\square$

This lemma, together with the well-known fact that $P(E^*) \leq P(E)$, or equivalently $\delta(E^*) \leq \delta(E)$, allows us to conclude, provided we can show that the quantitative isoperimetric inequality holds for axially symmetric sets in every dimension.

This was first proved by Hall (see [32]), though numerous alternative proofs have been discovered in recent years. Here, we will present the proof given by Maggi in [37]. This approach is particularly interesting because it relies on the Knothe map, much like Gromov's celebrated proof of the isoperimetric inequality.

Proof Sketch. By density in energy of smooth sets in finite perimeter sets, we can assume that E is bounded, axially symmetric, n -symmetric, with smooth boundary: indeed, all these properties are preserved by convolution with a radial kernel and they are inherited by level sets.

Moreover, thanks to the sharp isoperimetric inequality and the usual concavity trick for $t^{\frac{n-1}{n}}$, E behaves like its greatest connected component (up to an additional bound on $\delta(E)$), so we can assume that it is connected. In particular its radius is described by an even function $r \in C(\mathbb{R}, \mathbb{R}^+)$ such that $\{r > 0\} = (-a, a)$ for some $a \in \mathbb{R}^+$ and $r \in C^\infty(-a, a)$.

For $t \in (-a, a)$ let $\tau(t) \in \mathbb{R}$ be the unique value such that

$$|E \cap \{x_n < t\}| = |B \cap \{x_n < \tau(t)\}|$$

and define also

$$T(x', t) = \left(\frac{\sqrt{1 - \tau(x')^2}}{|x'|} x', \tau(t) \right)$$

for $(x', t) \in E$. This map is a smooth diffeomorphism between E and B and it satisfies $\det T = 1$, therefore, analogously to the computation we perform in the next section, we get

$$\begin{aligned} P(E) - P(B) &\geq \int_E (\operatorname{div} T - n) dx = \int_{-a}^a \frac{v_E(t)}{\tau'(t)^{\frac{1}{n-1}}} \left(n - 1 + \tau'(t)^{\frac{n}{n-1}} - n\tau'(t)^{\frac{1}{n-1}} \right) dt \\ &\geq \frac{n}{2} \int_{-a}^a \frac{v_E(t)}{\tau'(t)^{\frac{1}{n-1}}} \left(\tau'(t)^{\frac{1}{n-1}} - 1 \right)^2 dt \geq \frac{\left(\int_{-a}^a v_E(t) |\tau'(t)^{\frac{1}{n-1}} - 1| dt \right)^2}{\int_{-a}^a v_E(t) \tau'(t)^{\frac{1}{n-1}} dt}. \end{aligned}$$

Finally one easily checks that the denominator is bounded, and in addition, by a change of variable and a Poincaré inequality (which actually requires some work, as the interval might not be uniformly bounded), we can obtain

$$|E\Delta B| \lesssim \int_{-a}^a |1 - \tau'(t)| dt.$$

Moreover it is possible, but non-trivial, to assume that τ' is bounded, so that $|\tau'(t) - 1| \asymp |\tau'(t)^{\frac{1}{n-1}} - 1|$, which concludes. $\square$

2.3 Optimal Transport Method

Here we present the approach carried out by Figalli, Maggi and Pratelli in [27], which is based on a brilliant computation involving the Brenier map T from $\frac{1}{|E|}\chi_E$ to $\frac{1}{|B|}\chi_B$ for $|E| = |B|$. Up to rescaling, we assume that B has radius 1.

Remark. An outstanding feature of their approach is that they were able to deal directly with the general anisotropic case, without major additional difficulties.

Let us first recall the following results.

Theorem 2.3.1 (Brenier (see [2])). *Given any two probability measures $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^n)$, i.e., with finite second moment, such that μ is absolutely continuous with respect to the Lebesgue measure, there exists a unique optimal transport map T for the cost $c(x, y) = |x - y|^2$ and there exists a convex function $\phi: \mathbb{R}^n \rightarrow \mathbb{R}$ such that $T = \nabla\phi$ μ -a.e.*

Theorem 2.3.2 (see [3]). *Every convex function $\phi: \mathbb{R}^n \rightarrow \mathbb{R}$ is locally Lipschitz. Moreover $\nabla^2\phi$ is a locally finite Radon measure valued in symmetric positive semi-definite matrices, and consequently $\nabla\phi \in BV_{loc}(\mathbb{R}^n; \mathbb{R}^n)$.*

It directly follows that $T \in BV_{loc}(\mathbb{R}^n; \mathbb{R}^n)$ and DT is valued in symmetric positive semi-definite matrices, so in particular $\text{Div}(T)$ is a positive Radon measure. Moreover, it is well known that $\det \nabla T = 1$ a.e. on E , as one would expect in a smooth setting (see [2]). Then

$$\begin{aligned} |E| &= \int_E dx = \int_E (\det \nabla T)^{\frac{1}{n}} dx \leq \frac{1}{n} \int_E \text{div}(T) dx \leq \frac{1}{n} \text{Div}(E^{(1)}) \\ &= \frac{1}{n} \int_{\partial^* E} \text{tr}_E(T) \cdot \nu_E d\mathcal{H}^{n-1} \leq \frac{1}{n} \int_{\partial^* E} d\mathcal{H}^{n-1} = \frac{1}{n} P(E), \end{aligned}$$

where we used $|T| \leq 1$ $\mathcal{L}^n$ -a.e. $\implies |\text{tr}_E(T)| \leq 1$ $\mathcal{H}^{n-1}$ -a.e.. Analyzing the intermediate steps gives the following inequalities:

$$\int_E \left(\frac{1}{n} \text{div} T - 1 \right) dx + \frac{1}{n} |D^s T|(E^{(1)}) \leq \frac{1}{n} \text{Div}(T)(E^{(1)}) - \omega_n \leq \omega_n \delta(E) \quad (2.3)$$

$$\int_{\partial^* E} (1 - |\text{tr}_E(T)|) d\mathcal{H}^{n-1} \leq \int_{\partial^* E} (1 - \text{tr}_E(T) \cdot \nu_E) d\mathcal{H}^{n-1} \leq n\omega_n \delta(E), \quad (2.4)$$

where we applied $\operatorname{Div}^s(T) \geq |D^s T|$.

They can be further exploited by applying the following quantitative version of the AM-GM inequality. Instead of using their lemma, we will apply a result by Alzer (see [1]) as Segal did in [38].

Lemma 2.3.1 (Alzer). *Given $0 < x_1 \leq \dots \leq x_n$ real numbers, their arithmetic and geometric means m_A and m_G satisfy*

$$2n(m_A - m_G) \geq \frac{1}{x_n} \sum_{i=1}^n (x_i - m_G)^2.$$

Remark. This inequality is actually sharp: for example the case $x_1 = \dots = x_{n-2} = 1$, $x_{n-1} = \frac{1}{1+\varepsilon}$, $x_n = 1 + \varepsilon$ gives

$$LHS = \frac{2\varepsilon^2}{(1+\varepsilon)} \sim 2\varepsilon^2 \quad RHS = \frac{2\varepsilon^2 + 2\varepsilon^3 + \varepsilon^4}{(1+\varepsilon)^3} \sim 2\varepsilon^2.$$

To be precise, this implies that

$$2n\omega_n\delta(E) \geq 2n \int_E \left(\frac{1}{n} \operatorname{div} T - 1 \right) dx \geq \int_E \frac{1}{\lambda_n} |\nabla T - \operatorname{Id}|^2 dx \geq \frac{1}{\|\lambda_n\|_1} \|\nabla T - \operatorname{Id}\|_1^2,$$

which, applied to $\lambda_n \leq 1 + |\nabla T - \operatorname{Id}|$, also gives

$$\|\lambda_n\|_1 \leq |K| + \sqrt{2n\omega_n \|\lambda_n\|_1} \implies \|\lambda_n\|_1 \leq 4n\omega_n$$

by solving the quadratic (assuming $\delta(E) \leq 1$). Putting these pieces together we get

$$\int_E |\nabla T - \operatorname{Id}| dx \leq \sqrt{8n\omega_n} \sqrt{\delta(E)},$$

and finally, since $|D^s T|(E^{(1)}) \leq n\omega_n\delta(E)$, the function $S = T - \operatorname{Id}$ satisfies

$$|DS|(E^{(1)}) \leq 4n\omega_n \sqrt{\delta(E)}. \quad (2.5)$$

At this point, one would like to have either a Poincaré or a trace inequality, in order to recover $|T - \operatorname{Id}|$. It turns out that the trace is better suited for bounding the asymmetry index, thanks to 2.7.

For well-behaved sets, it holds in the following form.

Definition 2.3.1. *Given a finite perimeter set E with $|E| < \infty$, let*

$$\tau(E) = \inf \left\{ \frac{\mathcal{P}(F)}{\mathcal{H}^{n-1}(\partial^* F \cap \partial^* E)} : F \subset E, 0 < |F| \leq \frac{|E|}{2} \right\}.$$

Note that, equivalently, for any such F we have

$$(\tau(E) - 1) \mathcal{H}^{n-1}(\partial^* F \cap \partial^* E) \leq \mathcal{H}^{n-1}(\partial^* F \cap E^{(1)})$$

and $\tau(E) \geq 1$ for every admissible E .

Theorem 2.3.3. *For every E and $f \in BV(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$ we have*

$$(\tau(E) - 1) \inf_{a \in \mathbb{R}} \int_{\partial^* E} \text{tr}_E(|f - a|) \leq |Df|(E^{(1)}).$$

Proof. Without loss of generality $|E \cap \{f < 0\}| \leq \frac{|E|}{2}$ and $|E \cap \{f \geq 0\}| \leq \frac{|E|}{2}$. Define $F_t = \{f > t\} \cap E$, then by the coarea formula and Fubini's theorem

$$\begin{aligned} |Df^+|(E^{(1)}) &= \int_{\mathbb{R}^+} \mathcal{H}^{n-1}(\partial^* \{f > t\} \cap E^{(1)}) dt \\ &= \int_{\mathbb{R}^+} \mathcal{H}^{n-1}(\partial^* F_t \cap E^{(1)}) dt \\ &\geq (\tau(E) - 1) \int_{\mathbb{R}^+} \mathcal{H}^{n-1}(\partial^* F_t \cap \partial^* E) dt \\ &\geq (\tau(E) - 1) \int_{\mathbb{R}^+} \mathcal{H}^{n-1}(\{\text{tr}_E(f) > t\} \cap \partial^* E) dt \\ &= (\tau(E) - 1) \int_{\partial^* E} \text{tr}_E(f_+) d\mathcal{H}^{n-1}, \end{aligned}$$

where the second step holds because inside $E^{(1)}$ intersecting with E does not affect densities, and the fourth one follows from

$$\{\text{tr}_E(f) > t\} \cap \partial^* E \subset \partial^* F_t \cap \partial^* E.$$

Finally, since the same clearly holds for f^- , summing these two inequalities gives the thesis. $\square$

Luckily, if we assume the isoperimetric deficit to be small, it is possible to remove a bad portion of our set E , in a way that makes the previous inequality non-trivial.

Lemma 2.3.2. *For any $\lambda > 1$, if the family*

$$\mathcal{F}_\lambda = \left\{ F \subset E : 0 < |F| \leq \frac{|E|}{2}, P(F) \leq \lambda \mathcal{H}^{n-1}(\partial^* F \cap \partial^* E) \right\}$$

is non-empty, then it admits a maximal element with respect to a.e. inclusion.

Proof. Simply construct by induction an increasing sequence of sets F_h such that, if $s_h = \sup\{E \in \mathcal{F}_\lambda : F_h \subset E\}$, $|F_{h+1}| \geq \frac{|F_h| + s_h}{2}$, and define $F_\infty = \bigcup_{h \in \mathbb{N}} F_h$. If $F_\infty \in \mathcal{F}_\lambda$, it is trivial to check that it is maximal, so we are left to prove that $F_\infty \in \mathcal{F}_\lambda$: if $x \in F_h^{(\frac{1}{2})} \cap E^{(\frac{1}{2})}$, then

$$\frac{1}{2} = \Theta(x, F_h) \leq \Theta(x, F_{h+1}) \leq \Theta(x, E) = \frac{1}{2},$$

so $x \in F_{h+1}^{(\frac{1}{2})}$ and the sequence $F_h^{(\frac{1}{2})} \cap E^{(\frac{1}{2})} \subset F_\infty^{(\frac{1}{2})} \cap E^{(\frac{1}{2})}$ is increasing, which proves that the inequality passes to the limit:

$$P(F_\infty) \leq \liminf_h P(F_h) \leq \lambda \lim_h \mathcal{H}^{n-1}(\partial^* F_h \cap \partial^* E) \leq \lambda \mathcal{H}^{n-1}(\partial^* F_\infty \cap \partial^* E).$$

$\square$

Now we are ready to employ this lemma to restrict to a set with $\tau > 1$ nicely bounded, but first, we need to understand the behavior of such subsets.

Let once again $\psi(t) = t^{\frac{n-1}{n}} + (1-t)^{\frac{n-1}{n}} - 1$, for which one easily sees that

$$\psi(t) \geq (2 - 2^{\frac{n-1}{n}})t^{\frac{n-1}{n}} \quad \forall t \in \left[0, \frac{1}{2}\right], \quad (2.6)$$

and from now on let $k(n) = \frac{2-2^{\frac{n-1}{n}}}{3}$ (which is $\leq \frac{1}{3}$ and asymptotic to $\frac{2\log(2)}{3}\frac{1}{n}$).

Lemma 2.3.3. *Given $\lambda = 1 + k(n)$ and $F \subset E$ such that*

$$0 < |F| \leq \frac{|E|}{2} \quad P(F) \leq \lambda \mathcal{H}^{n-1}(\partial^* E \cap \partial^* F),$$

we have

$$P(E \setminus F) \leq P(E) \quad |F| \leq \left(\frac{\delta(E)}{k(n)}\right)^{\frac{n}{n-1}} |E|.$$

If in addition $\delta(E) \leq k(n)$, we also get $\delta(E \setminus F) \leq \frac{3}{k(n)}\delta(E)$.

Proof. Note that

$$\begin{aligned} P(E \setminus F) &= \mathcal{H}^{n-1}(\partial^* E \setminus \partial^* F) + \mathcal{H}^{n-1}(\partial^* F \cap E^{(1)}) \\ &\leq \mathcal{H}^{n-1}(\partial^* E \setminus \partial^* F) + k(n)\mathcal{H}^{n-1}(\partial^* E \cap \partial^* F) \\ &= P(E) - (1 - k(n))\mathcal{H}^{n-1}(\partial^* E \cap \partial^* F) \leq P(E) - (1 - k(n)^2)P(F), \end{aligned}$$

so the first property is proven. For the second one, let $t = \frac{|F|}{|E|}$, then by the sharp isoperimetric inequality

$$P(E) \geq P(E \setminus F) + (1 - k(n)^2)P(F) \geq n\omega_n^{\frac{1}{n}}((1-t)^{\frac{n-1}{n}} + (1 - k(n)^2)t^{\frac{n-1}{n}}),$$

or equivalently

$$\delta(E) \geq \psi(t) - k(n)^2 t^{\frac{n-1}{n}} \geq k(n)t^{\frac{n-1}{n}},$$

which is precisely the second property we wanted to prove.

Finally

$$\delta(E \setminus F) \leq \frac{P(E)}{n\omega_n(1-t)^{\frac{n-1}{n}}} - 1 \leq \frac{P(E)}{n\omega_n}(1+2t) - 1 = \delta(E) + 2t(\delta(E) + 1) \leq \frac{3}{k(n)}\delta(E).$$

□

Theorem 2.3.4. *If $\delta(E) \leq \frac{k(n)^2}{8}$, then there exists $G \subset E$ such that*

$$|E \setminus G| \leq \frac{\delta(E)}{k(n)}|E| \quad \delta(G) \leq \frac{3}{k(n)}\delta(E) \quad \tau(G) \geq 1 + k(n).$$

Proof. Let F_∞ be the set given by lemma 2.3.2 for $\lambda = 1 + k(n)$ (if $\mathcal{F}_\lambda$ is empty simply put $F_\infty = \emptyset$) and define $G = E \setminus F_\infty$. The first two properties trivially follow from the previous lemma. For the third property, if there were a contradicting set $H \subset G$, the hypotheses would imply that $H \cup F_\infty \in \mathcal{F}_\lambda$, contradicting the maximality of F_∞ : indeed,

$$\begin{aligned} |H \cup F_\infty| &= |H| + |F_\infty| \leq \frac{\delta(G)}{k(n)}|G| + \frac{\delta(E)}{k(n)}|E| \\ &\leq 3\frac{\delta(E)}{k(n)^2}|E| + \frac{\delta(E)}{k(n)}|E| < 4\frac{\delta(E)}{k(n)^2}|E| \leq \frac{|E|}{2}, \end{aligned}$$

where we applied the previous lemma to both H and G .

For the other property, let us split these perimeters in nicer pieces: let

$$\begin{aligned} a_1 &= \mathcal{H}^{n-1}(\partial^* H \cap G^{(1)}) \\ a_2 &= \mathcal{H}^{n-1}(\partial^* H \cap \partial^* G \cap \partial^* E) = \mathcal{H}^{n-1}(\partial^* H \cap \partial^* E) \\ a_3 &= \mathcal{H}^{n-1}(\partial^* H \cap \partial^* G \cap E^{(1)}) \\ a_4 &= \mathcal{H}^{n-1}(\partial^* F_\infty \cap E^{(1)}) - a_3 = \mathcal{H}^{n-2}(\partial^* F \cap E^{(1)} \setminus \partial^* H) \\ a_5 &= \mathcal{H}^{n-1}(\partial^* F_\infty \cap \partial^* E). \end{aligned}$$

Then the inequalities satisfied by H and F_∞ read as

$$a_1 \leq k(n)(a_2 + a_3) \quad a_3 + a_4 \leq k(n)a_5$$

and their sum gives

$$(a_1 + a_4) + a_3 \leq k(n)(a_2 + a_5) + k(n)a_3 \implies a_1 + a_4 \leq k(n)(a_2 + a_5),$$

which reads as

$$\mathcal{H}^{n-1}(\partial^*(H \cup F_\infty) \cap E^{(1)}) \leq k(n)\mathcal{H}^{n-1}(\partial^*(H \cup F_\infty) \cap \partial^* E).$$

□

In order to conclude, we need a statement which relates our integrals to the asymmetry index.

Lemma 2.3.4. *For every set $E \subset \mathbb{R}^n$ of finite perimeter with $|E| < \infty$ we have*

$$\int_{\partial^* E} |1 - |x|| \, d\mathcal{H}^{n-1} \geq |E \setminus B|. \quad (2.7)$$

Proof. Up to rescaling B , we can assume that $\mathcal{H}^{n-1}(\partial^* E \cap \partial B) = 0$, then

$$\nu_{E \setminus B} \mathcal{H}^{n-1} \llcorner \partial^*(E \setminus B) = \nu_E \mathcal{H}^{n-1} \llcorner (\partial^* E \setminus B) - \nu_B \mathcal{H}^{n-1} \llcorner (\partial B \cap E^{(1)}).$$

Now we are ready for the following computation

$$\begin{aligned} \int_{\partial^* E \setminus B} |x| - 1 &\geq \int_{\partial^* E \setminus B} (|x| - 1) \frac{x}{|x|} \cdot \nu_E \\ &= \int_{\partial^*(E \setminus B)} (|x| - 1) \frac{x}{|x|} \cdot \nu_{E \setminus B} = \int_{E \setminus B} \operatorname{div} \left((|x| - 1) \frac{x}{|x|} \right) \geq |E \setminus B|, \end{aligned}$$

where the second step holds because $|x| - 1 = 0$ on ∂B and the last one holds because the integrand is ≥ 1 . $\square$

Next, let G be the set given by lemma 2.3.4, then rescale G and E by the same factor so that $|G| = |B| = \omega_n$, and let T be the transport map from G to B . If we apply the trace inequality to the components of $S = T - \operatorname{Id}$ (up to a translation), together with

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n |x_i| \leq |x| \leq \sum_{i=1}^n |x_i| \quad \forall x \in \mathbb{R}^n,$$

we get

$$4n\omega_n \sqrt{\delta(G)} \geq |DS|(G^{(1)}) \geq \frac{k(n)}{\sqrt{n}} \int_{\partial^* G} |\operatorname{tr}_G(T) - \operatorname{Id}|.$$

If we add it to 2.4, and use that $|1 - |x|| \leq |S(x)| + (1 - |T(x)|)$, we get

$$\frac{5n^{\frac{3}{2}}}{k(n)} \omega_n \sqrt{\delta(G)} \geq \int_{\partial^* G} |1 - |x|| d\mathcal{H}^{n-1} \geq |G \setminus B| \geq \frac{\omega_n}{2} \alpha(G),$$

which means that

$$\alpha(G) \leq \frac{10n^{\frac{3}{2}}}{k(n)} \sqrt{\delta(G)}.$$

Now we just need to substitute E with G above: let $B(x)$ be a minimizing ball for $\alpha(G)$ and $r = \left(\frac{|E|}{|G|} \right)^{\frac{1}{n}}$, then

$$|E| \alpha(E) \leq |E \Delta (x + rB)| \leq |E \Delta G| + |G \Delta B(x)| + |B \Delta B(x)| = 2|E \setminus G| + \omega_n \alpha(G),$$

and in turn

$$\alpha(E) \leq \frac{2}{k(n)} \delta(E) + \frac{|G|}{|E|} \frac{10\sqrt{3}n^{\frac{3}{2}}}{k(n)^{\frac{3}{2}}} \sqrt{\delta(E)} \leq \frac{91n^{\frac{3}{2}}}{(2 - 2^{\frac{n-1}{n}})^{\frac{3}{2}}} \sqrt{\delta(E)},$$

which concludes the proof. Note that the constant is asymptotic to $\frac{273\sqrt{3}}{(2\log(2))^{\frac{3}{2}}} n^3$.

2.4 Selection Principle

The final general-dimensional approach we present is the selection principle, developed by Cicalese and Leonardi in [18]. Their idea is to consider a contradicting sequence, and substitute its elements with more regular sets with comparable

isoperimetric ratio, so that eventually they will contradict Fuglede's estimate, being graphs over the ball. We follow the exposition by Fusco (see [29]).

The proof will heavily rely on the following regular convergence result for almost minimizers by Tamanini (see [41]).

Theorem 2.4.1 (Tamanini). *If E is a (Λ, r_0) -almost-minimizer, then $\partial^* E$ is a $C^{1, \frac{1}{2}}$ manifold, $\partial E \setminus \partial^* E \subset \partial E$ is relatively closed and $\dim_{\mathcal{H}}(\partial E \setminus \partial^* E) \leq n - 8$. Moreover, if $E_j \rightarrow E$ is a uniformly bounded sequence of (Λ, r_0) -almost-minimizers and E is C^2 , then E_j is eventually $C^{1, \frac{1}{2}}$ and the convergence is $C^{1, \frac{1}{2}}$.*

Remark. Explicit constants in this result are heavily dependent on the dimension n , and there is no hope for them to be universally small enough to be useful for our purposes.

First, by lemma 2.2.1 we can restrict ourselves to uniformly bounded sets $E \subset B_R$ for some $R(n) > 0$; then it is sufficient to prove the following.

Theorem 2.4.2. *There exists $\delta(n) > 0$ such that if $E \subset B_R$, $|E| = \omega_n$ and $\delta(E) \leq \delta(n)$, then*

$$\alpha(E) \leq \sqrt{6n} \sqrt{\delta(E)}.$$

Proof. By contradiction, let $E_j \subset B_R$ be a sequence such that $\frac{\alpha(E_j)}{\sqrt{\delta(E_j)}} \geq \sqrt{6n}$ and $\delta(E_j) \rightarrow 0$. Then one can modify the sequence as follows: let F_j be a minimizer of the problem

$$\min\{P(F) + \omega_n |\alpha(F) - \alpha(E_j)| + \Lambda ||F| - \omega_n| : F \subset B_R\}$$

for a constant $\Lambda > 0$ to be chosen later, which exists by BV compactness (here it is relevant that we restricted to B_R). These newly defined sets F_j satisfy a few nice properties:

1. If $\Lambda > n$, then $F_j \rightarrow B$: in fact, up to a subsequence, F_j converges to some $F \subset B_R$ which minimizes

$$\min\{P(E) + \omega_n \alpha(E) + \Lambda ||E| - \omega_n| : E \subset B_R\},$$

and it is easy to check that the only minimizers are the unitary balls.

2. There exists a $\lambda > 0$ such that all the F_j are (λ, R) -almost-minimizers: let $F_j \Delta G \Subset B_r(x)$, then we shall distinguish two cases:

- $G \subset B_R$: by the minimality of F_j and the triangle inequality

$$\begin{aligned} P(F_j) &\leq P(G) + \omega_n |\alpha(G) - \alpha(F_j)| + \Lambda ||G| - |F_j|| \\ &\leq P(G) + (\Lambda + 1) |G \Delta F_j| \leq P(G) + (\Lambda + 1) \omega_n r^n. \end{aligned}$$

- $|G \setminus B_R| > 0$:

$$P(F_j) - P(G) = (P(F_j) - P(G \cap B_R)) + (P(G \cap B_R) - P(G)),$$

where the first term is less than $\Lambda\omega_n r^n$ as above, while for the second one

$$P(G \cap B_R) - P(G) = P(B_R) - P(G \cup B_R) \leq 0$$

by the sharp isoperimetric inequality.

3. From the definition of E_j and F_j we get

$$\begin{aligned} P(F_j) + \Lambda||F_j| - \omega_n| + \omega_n|\alpha(F_j) - \alpha(E_j)| &\leq P(E_j) \\ &\leq P(B) + \frac{\omega_n}{6}\alpha(E_j)^2 \leq P(F_j) + \Lambda||F_j| - \omega_n| + \frac{\omega_n}{6}\alpha(E_j)^2, \end{aligned}$$

and in particular $|\alpha(F_j) - \alpha(E_j)| \leq \frac{1}{6}\alpha(E_j)^2$, which implies that $\frac{\alpha(F_j)}{\alpha(E_j)} \rightarrow 1$.

Furthermore, let us rescale $F'_j = r_j F_j$ so that $|F'_j| = \omega_n$, and also translate so that their barycenters are in the origin. Tamanini's result ensures that the sets F_j are eventually graphical and also satisfy the hypotheses of Fuglede's estimate. Then by the C^1 convergence we have $r_j \rightarrow 1$ and $P(F_j) \rightarrow n\omega_n$, hence eventually $P(F_j) < \Lambda|F_j|$.

Before concluding, further note that

$$P(F'_j) - P(F_j) = P(F_j)|r_j^{n-1} - 1| \leq \Lambda|F_j||r_j^n - 1| = \Lambda||F'_j| - |F_j|| = \Lambda||F_j| - \omega_n|,$$

where the last equality holds because F_j is convex. Finally we can compute

$$P(F'_j) \leq P(F_j) + \Lambda||F_j| - \omega_n| \leq P(B) + \frac{\omega_n}{6}\alpha(E_j)^2,$$

and the conclusion clearly follows from observing that $\frac{\alpha(F'_j)}{\alpha(E_j)} \rightarrow 1$. □

Let us finally remark that, when $n = 2$, this proof actually yields a sharper estimate, based on the following result for convex sets (see [33]).

Theorem 2.4.3 (Hall, Hayman). *Every convex set E satisfies*

$$\delta(E) \geq \frac{\pi}{8(4-\pi)}\alpha(E)^2 + o(\alpha(E)^2),$$

and $C = \frac{\pi}{8(4-\pi)}$ is the best such constant.

The proof, as often happens in two dimensions, consists in performing a fine analysis of the Fourier coefficients of the boundary function, seen as a periodic complex valued curve.

In [18] the authors observed that, thanks to the selection principle, this result easily extends to all sets, for the following reason: since the new sequence F_n is eventually converging to B in $C^{1,\alpha}$, the sets are eventually convex.

Chapter 3

Further Isoperimetry

3.1 Finer Estimates

We begin by demonstrating how the selection principle of Cicalese and Leonardi has been successfully applied in other contexts, specifically by presenting the quantitative isoperimetric inequality of Fusco and Julin (see [30]). They proved that one can substitute the asymmetry index α with a finer candidate, namely

$$\beta(E) = \sqrt{\min_{y \in \mathbb{R}^n} \int_{\partial^* E} \left| \nu_E(x) - \frac{x - y}{|x - y|} \right|^2 d\mathcal{H}^{n-1}(x)}.$$

This quantity better captures the oscillations of the boundary, and in fact the following holds.

Lemma 3.1.1. *If $|E| = |B(r)|$, then*

$$\alpha(E)^2 + \delta(E) \lesssim r^{1-n} \beta(E)^2.$$

This bound follows from the identity

$$\beta(E)^2 = P(E) - (n-1) \max_{y \in \mathbb{R}^n} \int_E \frac{1}{|x - y|} dx \quad (3.1)$$

and a simple concavity argument.

Let us examine how the proof of quantitative stability must be modified in this context. The appropriate selection principle will not immediately give us almost minimizers, but only quasi minimizers.

Definition 3.1.1. *A set $E \subset \mathbb{R}^n$ is called (K, r_0) -quasi-minimizer if for every $F \subset \mathbb{R}^n$ such that $E \Delta F \subseteq B_r(x)$ for some $r < r_0$, we have*

$$P(E, B_r(x)) \leq KP(F, B_r(x)).$$

For this reason, we will need the following lemma.

Theorem 3.1.1 (David, Semmes (see [21])). *If E is a (K, r_0) -quasi-minimizer, there exists a constant $C > 0$ such that for every $x \in \partial^* E$ and $0 < r < r_0$ there exist $y, z \in B_r(x)$ satisfying*

$$B\left(y, \frac{r}{C}\right) \subset E \quad B\left(z, \frac{r}{C}\right) \cap E = \emptyset.$$

Now we are ready to prove that there exists $C(n) > 0$ such that

$$\beta(E) \leq C(n) \sqrt{\delta(E)}.$$

Proof Sketch. First, we need to provide an appropriate Fuglede-type estimate, which follows directly from the proof of the original result. Indeed, let E be a nearly spherical set, then using the identity $|\nu - \mu|^2 = 2(1 - \nu \cdot \mu)$ for any unit vectors $\nu, \mu \in \mathbb{S}^{n-1}$, we can bound the asymmetry as follows:

$$\begin{aligned} \beta(E)^2 &\leq \frac{1}{2} \int_{\mathbb{S}^{n-1}} \left(1 - \nu_E \cdot \frac{x}{|x|} \right) d\sigma \\ &\lesssim \int_{\mathbb{S}^{n-1}} \left(1 - \frac{1+u}{\sqrt{(1+u)^2 + |\nabla u|^2}} \right) d\sigma \\ &\lesssim \int_{\mathbb{S}^{n-1}} |\nabla u|^2 d\sigma \lesssim \delta(E), \end{aligned}$$

where the second step follows from the area formula, and the final inequality is an intermediate byproduct in the proof of the original Fuglede estimate.

Returning to the selection principle: as before, by Lemma 2.2.1, we may assume $E \subset B_R$. We then take a contradicting sequence $E_j \subset B_R$ with $|E_j| = \omega_n$. This time we consider the minimization problem

$$\min \left\{ \mathcal{F}(F) = P(F) + \frac{1}{4} |\beta(F)^2 - \beta(E_j)^2| + \Lambda ||F| - \omega_n| : F \subset B_R \right\}$$

for some $\Lambda > n$. This functional turns out to be lower semi-continuous, therefore minimizers F_j exist and up to a subsequence $F_j \rightarrow F_\infty$ where F_∞ is a solution of

$$\{P(F) + \Lambda ||F| - \omega_n| + \frac{1}{4} \beta(F)^2\},$$

and in turn $F_\infty = B(x_0)$ for some $x_0 \in \mathbb{R}^n$.

Analogous to the Cicalese-Leonardi approach, the minimality of F_j implies (K, r_0) -quasi-minimality for some uniform (K, r_0) . Furthermore, the preceding lemma implies that for every $\varepsilon > 0$

$$B(x_0, 1 - \varepsilon) \subset F_j \subset B(x_0, 1 + \varepsilon)$$

eventually in j . Indeed, the lemma asserts that F_j has both some mass and some holes close to the boundary, so ∂F_j cannot lie far from ∂B .

Finally, up to a translation assume that the F_j satisfy $|F_j| = \omega_n$ and are centered in the origin. Then the minimality, together with the identity 3.1, implies that the F_j are also (Λ', r_0) minimizers for some $\Lambda' > 0$, and the conclusion follows from a simple computation analogous to Cicalese-Leonardi. $\square$

In the same spirit, Bögelein, Duzaar and Fusco were able to obtain a quantitative isoperimetric inequality on $\mathbb{S}^n$ (see [9]) and, most importantly, for sets of higher codimension, represented by currents (see [8]). In both cases, the proof required an ad hoc Fuglede-type estimate and a $C^{1,\alpha}$ convergence result for Λ -minimizers.

Lastly, we mention that the same approach turned out to be extremely useful even in non-local settings. The most relevant instance for us is the quantitative stability for s -perimeters with $s \in (0, 1)$ by Figalli, Fusco, Maggi, Millot and Morini (see [26]).

One of the most interesting features of this result is the following: since the s -perimeter converges to the usual perimeter for $s \nearrow 1$, and their estimate is uniform for $s > \delta > 0$, it also implies the usual quantitative isoperimetric inequality.

3.2 The Gaussian Case

Another important setting to mention is the Gaussian one. In this context, the optimal asymptotic growth has been completely determined by Barchiesi, Brancolini, and Julin [5].

For a locally finite perimeter set $E \subset \mathbb{R}^n$, let

$$\gamma = \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{|x|^2}{2}} \cdot \mathcal{L}^n$$

be the Gaussian measure in $\mathbb{R}^n$ and $b(E) = \int_E x \, d\gamma$ be the non-normalized Gaussian barycenter. We define the strong asymmetry index as

$$\beta_\gamma(E) = \min_{\nu \in \mathbb{S}^{n-1}} |b(E) - b(H_{\nu,s})|,$$

where $H_{\nu,s} = \{x \in \mathbb{R}^n : x \cdot \nu > s\}$ is any half-space such that

$$\Phi(s) = \frac{1}{\sqrt{2\pi}} \int_s^\infty e^{-\frac{t^2}{2}} dt = \gamma(H_{\nu,s}) = \gamma(E).$$

Moreover the (rescaled) Gaussian perimeter of a finite perimeter set is defined as

$$P_\gamma(E) = \frac{1}{(2\pi)^{\frac{n-1}{2}}} \int_{\partial^* E} e^{-\frac{|x|^2}{2}} d\mathcal{H}^{n-1}(x),$$

or equivalently

$$P_\gamma(E) = \sqrt{2\pi} \sup \left\{ \int_E (\operatorname{div} \varphi - x \cdot \varphi) : \varphi \in C_c^\infty(\mathbb{R}^n; \mathbb{R}^n), \|\varphi\|_\infty \leq 1 \right\}.$$

The sharp Gaussian isoperimetric inequality was proven independently by Borell (see [10]) and by Sudakov and Tsirelson (see [39]). It states that for any fixed volume $\Phi(s)$ the minimal perimeter is $P_\gamma(H_{\nu,s}) = e^{-\frac{s^2}{2}}$. More recently Carlen and Kerce proved that the half-spaces $H_{\nu,s}$ are the unique minimizers (see [15]).

In view of these results, we define the Gaussian isoperimetric deficit as

$$D_\gamma(E) = P(E) - e^{-\frac{s^2}{2}}.$$

The asymmetry index β_γ is a very good candidate here for two main reasons:

- It bounds the standard Fraenkel asymmetry $\alpha_\gamma(E) = \min_{\nu \in \mathbb{S}^{n-1}} \gamma(E \Delta H_{\nu,s})$ as follows:

$$\frac{e^{\frac{s^2}{2}}}{4} \alpha_\gamma(E)^2 \leq \beta_\gamma(E).$$

- It is algebraically easier to handle and it is related to the strong asymmetry β , in fact it satisfies

$$2D_\gamma(E) + 2\sqrt{2\pi}\beta_\gamma(E) = \frac{1}{(2\pi)^{\frac{n-1}{2}}} \min_{\nu \in \mathbb{S}^{n-1}} \int_{\partial^* E} |\nu_E - \nu|^2 d\mathcal{H}^{n-1} = \mathcal{E}(E),$$

where $\mathcal{E}$ is the excess.

We expect a dimension-free estimate because the Gaussian setting is inherently dimension-free, which is naturally reflected in the fact that many of its functional inequalities hold with universal constants.

Theorem 3.2.1 (Barchiesi, Brancolini, Julin). *There exists a universal constant $C > 0$ such that for every $s \in \mathbb{R}$ and $E \subset \mathbb{R}^n$ with $\gamma(E) = \Phi(s)$, we have*

$$\beta_\gamma(E) \leq C(1 + s^2)D_\gamma(E).$$

The proof outline might look similar to a selection principle, but it is actually a finer argument: consider the functional

$$\mathcal{F}(E) = P_\gamma(E) + \frac{\varepsilon}{2}|b(E)|^2 + \Lambda|\gamma(E) - \Phi(s)|$$

and most importantly its minimizers. First let us mention that we are interested in a fixed large Λ , so that the last term is almost a volume constraint $\gamma(E) = \Phi(s)$, but enjoys the algebraic properties of a volume penalization.

Moreover note that, for a fixed volume, half-spaces are the minimizers of P_γ , but on the other hand they are also the maximizers of $|b|$. Therefore we hope to

prove that, if $\varepsilon > 0$ is universally small enough, the minimizers of $\mathcal{F}$ are precisely the half-spaces. In fact, if this was the case, a simple comparison would imply our desired quantitative stability.

The strategy to prove this claim is the following: fix a minimizer of $\mathcal{F}$, which will be regular by Tamanini's theorem, and perform clever second variation arguments to prove that the inner normal ν_E must be aligned with the barycenter. This means that E is actually one-dimensional, namely $E = \mathbb{R}^{n-1} \times F$, and one just needs to check that F is a half-line.

Finally, it is worth noting that, while the only global minimizers for the Gaussian perimeter are the half-spaces, La Manna proved in [35] that balls are strict local minimizers among centrally symmetric sets.

3.3 On Manifolds

Lastly, we mention what is currently known about general manifolds.

As one might expect, a global quantitative isoperimetric inequality generally fails on manifolds because the underlying geometry strongly influences the stability of sets. However, Chodosh, Engelstein and Spolaor proved in [17] that generically a quantitative inequality with the Euclidean sharp exponent $\frac{1}{2}$ holds.

For a given Riemannian manifold (M, g) , define the isoperimetric profile

$$I_g(V) = \inf \{P_g(E) : E \subset M, |E|_g = V\}$$

the set of minimizers $\mathcal{M}_g^{|E|}$ and the asymmetry index

$$\alpha_g(E) = \inf \left\{ |E \Delta F|_g : F \in \mathcal{M}_g^{|E|} \right\}.$$

On the negative side, they found the following example.

Theorem 3.3.1 (Chodosh, Engelstein, Spolaor). *There exists a closed Riemannian manifold (M, g) such that for some $V > 0$ there exists a unique isoperimetric region $\Omega \subset M$ and smooth sets E_k such that $|\Omega \Delta E|_g \rightarrow 0$ and*

$$\frac{|\Omega \Delta E_k|_g^{2+\gamma}}{P_g(E_k) - P_g(\Omega)} \rightarrow \infty$$

for all $\gamma > 0$.

The construction is based on the well-known fact that for $R > 0$ big enough the manifold $\mathbb{S}^1(R) \times \mathbb{S}^{n-1}$ is such that the only half-volume isoperimetric regions are described as $(z, z + R\pi) \times \mathbb{S}^{n-1}$.

Next, let us focus on the positive results. For a given manifold let Γ be the set of C^3 metric endowed with the C^3 topology.

Theorem 3.3.2 (Chodosh, Engelstein, Spolaor). *Let M be a closed manifold of dimension $2 \leq n \leq 7$, then there exists a dense subset $\mathcal{G} \subset \Gamma$ such that each $g \in \mathcal{G}$ has the following property: there exists a dense subset $\mathcal{V} \subset (0, |M|_g)$ such that for each $V \in \mathcal{V}$ we have*

$$P_g(E) - I_g(V) \geq C(g, V)\alpha_g(E)^2$$

for any $E \subset M$ with $|E| = V$.

This strategy represents a standard approach in such scenarios: fix a diffeomorphism type of the boundary, define a suitable map that associates to each such minimizer its relative metric and volume. Then, after verifying that it is a Fredholm operator, we can apply the Sard-Smale theorem to prove that non-degeneracy (equivalent to strict stability) generally holds.

Finally let us mention that a weak, but sharp, stability result holds for any analytic metric.

Theorem 3.3.3 (Chodosh, Engelstein, Spolaor). *Let (M, g) be a closed analytic Riemannian manifold of dimension $2 \leq n \leq 7$, then for every volume $V > 0$ there exist $C > 0$ and $\gamma \geq 0$ such that*

$$P_g(E) - I_g(V) \geq C\alpha_g(E)^{2+\gamma}.$$

In this case the idea is to employ a specific Łojasiewicz-Simon inequality, borrowed from [25], asserting that analytic functions are somewhat strictly stable around minimizers, but only for some exponent greater than 2. In our setting we can apply this estimate to P_g around isoperimetric regions and get stability for general sets by a simple compactness argument.

Remark. In both statements, the dimension is bounded from above. This is unavoidable, because for $n \geq 8$ isoperimetric regions are not necessarily smooth, while the needed local stability results only work for smooth minimizers.

Remark. The reason why these estimates were so difficult to achieve is the following: in arbitrary Riemannian manifolds isoperimetric regions can be completely unpredictable and hard to describe, while the original Fuglede estimate strongly relies on the precise shape of balls.

This discussion should also convince the reader that these methods are extremely general, and consequently we should expect to be able to provide strong generic stability and weak analytic stability in many other scenarios.

Chapter 4

The Sobolev Case

Recall that, in a given dimension $n \geq 3$, the sharp Sobolev inequality states that for any function $f \in \dot{H}^1(\mathbb{R}^n)$

$$S_n \|f\|_{L^{2^*}(\mathbb{R}^n)}^2 \leq \|\nabla f\|_{L^2(\mathbb{R}^n)}^2 ,$$

where $2^* = \frac{2n}{n-2} = 2 + \frac{4}{n-2}$ is the Sobolev exponent and $S_n = \frac{1}{4}n(n-2)|\mathbb{S}^n|^{\frac{2}{n}}$. Moreover equality is only attained by the Aubin-Talenti bubbles

$$g_{a,b,y} = b \frac{1}{(1 + a|x - y|^2)^{\frac{n-2}{2}}}$$

for $a > 0$, $b \in \mathbb{R}$ and $y \in \mathbb{R}^n$. This was first proven independently by Aubin and Talenti in 1976 (see [4], [40]) and later by Carlen and Loss via a competing symmetry argument (see [16]).

Let $\mathcal{M}$ be the set of the Aubin-Talenti bubbles and $\mathcal{M}_1 \subset \mathcal{M}$ the bubbles with unitary L^{2^*} norm. A quantitative counterpart to this inequality that has recently gained significant interest is the following:

$$d_{\dot{H}^1}(f, \mathcal{M})^2 = \inf_{g \in \mathcal{M}} \|\nabla f - \nabla g\|_{L^2(\mathbb{R}^n)}^2 \leq C(n) \left(\|\nabla f\|_{L^2(\mathbb{R}^n)}^2 - S_n \|f\|_{L^{2^*}(\mathbb{R}^n)}^2 \right) . \quad (4.1)$$

The first full proof of the existence of a $C(n) > 0$ was found by Bianchi and Egnell in [7], but they did not provide explicit control over the constant due to the presence of compactness arguments.

Define also the ratio

$$\mathcal{E}(f) = \frac{\|\nabla f\|_{L^2(\mathbb{R}^n)}^2 - S_n \|f\|_{L^{2^*}(\mathbb{R}^n)}^2}{\inf_{g \in \mathcal{M}} \|\nabla f - \nabla g\|_{L^2(\mathbb{R}^n)}^2} ,$$

so that equivalently we want to prove that

$$\inf_{f \in \dot{H}^1(\mathbb{R}^n)} \mathcal{E}(f) \gtrsim \frac{1}{n} .$$

Lemma 4.0.1. *The functional $\mathcal{E}$ is conformally invariant in the following sense: given any $f \in \dot{H}^1(\mathbb{R}^n)$ and conformal transformation C , the function*

$$f_C(x) = |JC(x)|^{\frac{1}{2^*}} f(C(x))$$

satisfies $\mathcal{E}(f) = \mathcal{E}(f_C)$.

Proof. We just need to consider translations, dilations, orthogonal maps and the inversion $x \mapsto \frac{x}{|x|^2}$ by Liouville's theorem. Then the thesis easily follows from observing that $\mathcal{M}$ is preserved by these transformations, so that both numerator and denominator are invariant. $\square$

More precisely, every $g \in \mathcal{M}$ can be described as $g = c(g_0)_C$ for some conformal transformation C and a constant $c \in \mathbb{R}$, where $(g_0)_C$ is defined as in the previous lemma and

$$g_0(x) = \frac{1}{(1 + |x|^2)^{\frac{n-2}{2}}}$$

Remark. $d(f, \mathcal{M})$ is somewhat analogous to the asymmetry $\beta(E)$ introduced in the previous chapter. The L^{2^*} distance would correspond to the Fraenkel asymmetry index $\alpha(E)$.

4.1 Reductions

The first step requires a change of perspective: using the stereographic projection, we can restate our inequality on $\mathbb{S}^n$. This transformation will turn out to be highly advantageous because it maps Sobolev minimizers into a convenient set of functions that includes the constants.

More explicitly, recall that the stereographic projection is the map $p: \mathbb{R}^n \rightarrow \mathbb{S}^n$ such that

$$p(x) = \frac{1}{1 + |x|^2} (2x, 1 - |x|^2),$$

which in addition satisfies $Jp = \left(\frac{2}{1+|x|^2}\right)^n$. Let us consider the transformation that, given a function $f \in H^1(\mathbb{R}^n)$, gives

$$F(\omega) = \left(\frac{1}{1 + \omega_{n+1}}\right)^{\frac{n-2}{2}} f(p^{-1}(\omega)),$$

and note that $1 + \omega_{n+1} = \frac{2}{1+|x|^2}$, so that the constant coincides with

$$(J(p^{-1}))^{1/2^*} = \left(\frac{1 + |x|^2}{2}\right)^{\frac{n-2}{2}}.$$

In this way, one can easily check that change of variable gives

$$\|F\|_{L^{2^*}(\mathbb{S}^n)} = \|f\|_{L^{2^*}(\mathbb{R}^n)} \quad \int_{\mathbb{S}^n} \left(|\nabla F|^2 + \frac{1}{4}n(n-2)|F|^2 \right) d\sigma = \int_{\mathbb{R}^n} |\nabla f|^2 dx ,$$

while the Aubin-Talenti bubbles become

$$G_{a,b,c}(\omega) = c(a + b \cdot \omega)^{-\frac{n-2}{2}}$$

with $a > 0$, $c \in \mathbb{R}$, $b \in \mathbb{R}^n$ such that $|b| < a$, so that inequality 4.1 in terms of F becomes

$$\begin{aligned} & \inf_{a,b,c} \left\{ \|\nabla F - \nabla G\|_{L^2(\sigma)}^2 + A \|F - G\|_{L^2(\sigma)}^2 \right\} \\ & \leq C(n) \left(\|\nabla F\|_{L^2(\sigma)}^2 + A \|F\|_{L^2(\sigma)}^2 - A \|F\|_{L^{2^*}(\sigma)}^2 \right) , \end{aligned}$$

where $A = \frac{1}{4}n(n-2)$ and the surface measure σ has been normalized.

Restricting to the case where the distance minimizer is $G \equiv 1$, and defining $r = F - 1$, a standard first variation argument yields

$$\int_{\mathbb{S}^n} r d\sigma = 0 \quad \int_{\mathbb{S}^n} \omega r d\sigma = 0 ,$$

and our inequality simplifies to

$$\|\nabla r\|_{L^2}^2 + A \|r\|_{L^2}^2 \leq C(n) (\|\nabla r\|_{L^2}^2 + A \|1 + r\|_{L^2}^2 - A \|1 + r\|_{L^{2^*}}^2) \quad (4.2)$$

for any such $r \in H^1(\mathbb{S}^n)$. Our aim is to prove that we can choose $C(n) \leq Cn$ for a universal constant C .

4.2 Local Case

Assume that $r \geq -1$ (corresponding to positive F and f) and that $\|r\|_{L^q} \leq \delta$, where $\delta > 0$ is a universal constant to be determined. We then decompose our function as follows:

$$r_1 = r \wedge \varepsilon_1 \quad r_2 = (r - \varepsilon_1)_+ \wedge (\varepsilon_2 - \varepsilon_1) \quad r_3 = (r - \varepsilon_2)_+ = r - r_2 - r_1 ,$$

where $0 < 2\varepsilon_1 < \varepsilon_2 < 1$ are universal constants to be decided.

Remark. For any fixed n , the existence of a $\delta(n)$ can be simply proven by spectral theory and some polynomial expansions for the L^{2^*} norm (recall that $2^* \leq 6$). However, the technique we are going to present requires $2^* \leq 3$, or equivalently $n \geq 6$. As we are mainly interested in the asymptotic properties of 4.1, we will give the cases $n = 3, 4, 5$ for granted and assume $n \geq 6$.

Lemma 4.2.1. *There exists a constant $C_{\varepsilon_1, \varepsilon_2}$ such that for any such r and any $2 \leq q \leq 3$ there exists a constant $C_{\varepsilon_1, \varepsilon_2}$ for which*

$$(1+r)^q \leq 1+qr + \left(\frac{1}{2}q(q-1) + 2\varepsilon_1\theta\right) r_1^2 + \left(\frac{1}{2}q(q-1) + C_{\varepsilon_1, \varepsilon_2}\theta\right) r_2^2 + \sum_{i \neq j} r_i r_j + (1+\varepsilon_2\theta)r_3^q,$$

where $\theta = q - 2$.

Proof Sketch. While the precise proof is somewhat technical, the underlying strategy is quite natural. First, note that

$$(1+r)^q \leq 1+qr + \frac{q(q-1)}{2}r^2 + \theta r_+^3$$

for all $r \in [-1, \infty)$, e.g. by bounding the Lagrange remainder.

Now, for $x \leq \varepsilon_2$ the estimate easily follows from appropriate weighted AM-QM inequalities.

For $x > \varepsilon_2$ we need to estimate $(1+r)^q - r^q$ keeping track of the factor θ , which makes a difference when q approaches 2. Let us split in cases:

- If $r_3 \leq C$ for a fixed $C > 1 + \varepsilon_2$, then

$$(1+r)^q - (1+r)^2 = (1+r)^2((1+r)^\theta - 1) \leq (1+r)^2 r \theta \leq 8C^3 \theta,$$

and similarly one gets

$$r_3^2 - r_3^q = r_3^2(1 - r_3^{q-2}) \leq \min\{r_3^2, 1 - r_3^{q-2}\} \lesssim \theta$$

for $r_3 \leq 1$, which concludes this case.

- If $r_3 \geq C$, then

$$\begin{aligned} (1+r)^q &= (1+\varepsilon_2 + r_3)^q = r_3^q \left(1 + \frac{1+\varepsilon_2}{r_3}\right)^q \\ &\leq r_3^q + qr_3^{q-1}(1+\varepsilon_2) + \frac{q(q-1)}{2}r_3^{q-2}(1+\varepsilon_2)^2 + \theta C^{q-3}(1+\varepsilon_2)^3, \end{aligned}$$

and the conclusion follows from the bounds

$$qr_3^{q-1} - 2r_3 \lesssim \theta r_3^q \quad \frac{q(q-1)}{2}r_3^{q-2} - 1 \lesssim \theta r_3^q,$$

which hold for C universally big enough (and are again obtained by splitting in different regimes).

□

By applying this lemma, together with $(1+x)^{\frac{2}{q}} \leq 1 + \frac{2}{q}x$, to our function with $q = 2^*$ (so that $\theta = \frac{4}{n-2} \leq 1$) we get

$$\begin{aligned} \|1+r\|_{L^q}^2 &\leq 1 + (q-1+\varepsilon_1\theta) \int_{\mathbb{S}^n} r_1^2 d\sigma \\ &\quad + (q-1+C_{\varepsilon_1,\varepsilon_2}\theta) \int_{\mathbb{S}^n} r_2^2 d\sigma + \frac{2}{q} \sum_{i \neq j} \int_{\mathbb{S}^n} r_i r_j d\sigma + \frac{2}{q} (1+\varepsilon_2\theta) \int_{\mathbb{S}^n} r_3^q d\sigma, \end{aligned}$$

so that, for a fixed (small enough) $\varepsilon_0 > 0$ the RHS of 4.2 is greater than

$$\varepsilon_0\theta(\|\nabla r\|_2^2 + A\|r\|_2^2) + I_1 + I_2 + I_3,$$

where (note that $A\theta = n$)

$$\begin{aligned} I_1 &= (1-\varepsilon_0\theta)\|\nabla r_1\|_2^2 - n(1+\varepsilon_0+\varepsilon_1)\|r_1\|_2^2 + n\sigma_0(\|r_2\|_2^2 + \|r_3\|_2^2) \\ I_2 &= (1-\varepsilon_0\theta)\|\nabla r_2\|_2^2 - n(1+\varepsilon_0+\sigma_0+C_{\varepsilon_1,\varepsilon_2})\|r_2\|_2^2 \\ I_3 &= (1-\varepsilon_0\theta)\|\nabla r_3\|_2^2 + A(1-(\sigma_0+\varepsilon_0)\theta)\|r_3\|_2^2 - \frac{2}{q}A(1+\varepsilon_2\theta)\|r_3\|_q^q. \end{aligned}$$

Remark. Note that an upper bound on ε_0 is needed because in the last step we got rid of $(1-\frac{2}{q}-\varepsilon_0\theta)\sum_{i \neq j} \int r_i r_j$ where $1-\frac{2}{q} = \frac{2}{d}$ (this observation is missing in [23]). Note also that this sum is positive by definition.

Now it remains to show that these terms are all non-negative for an appropriate universal choice of the constants. Let us first give an outline of the proof:

1. I_1 : Similarly to Fuglede's approach, r_1 is a good approximation of r , so its degree 0 and 1 harmonic coefficients are almost null, and in turn it satisfies a strong Poincaré inequality.
2. I_2 : We expect r_2 to behave like a constant, so that also higher degree harmonic coefficients are almost vanishing, and in turn it would satisfy an arbitrarily strong Poincaré inequality (needed to balance the magnitude of $C_{\varepsilon_1,\varepsilon_2}$).
3. I_3 : Note that $\|r_3\|_q \leq \|r\|_q \leq 1$ implies $\|r_3\|_q^q \leq \|r_3\|_q^2$. Then, since $\frac{2}{q} = \frac{n-2}{n} = 1 - \frac{2}{n}$, it is clear that there exists a universal constant $C > 0$ such that $I_3 \geq 0$ is weaker than the sharp Sobolev inequality, whenever $\max\{\varepsilon_0, \sigma_0, \varepsilon_2\} \leq C$.

$$A\|r_3\|_q^2 \leq \|\nabla r_3\|_2^2 + A\|r_3\|_2^2.$$

4.2.1 First term

Let

$$\tilde{r} = r_1 - \int_{\mathbb{S}^n} r_1 d\sigma - (n+1)\omega \cdot \int_{\mathbb{S}^n} \omega r_1 d\sigma$$

be the projection of r_1 on the spherical harmonics of degree ≥ 2 , then we know that

$$\begin{aligned} \|r_1\|_2^2 - \left(\int_{\mathbb{S}^n} r_1 d\sigma \right)^2 - (n+1) \left| \int_{\mathbb{S}^n} \omega r_1 d\sigma \right|^2 &= \|\tilde{r}\|_{L^2}^2 \\ &\leq \frac{1}{2(n+1)} \|\nabla \tilde{r}\|_{L^2}^2 \leq \frac{1}{2(n+1)} \|\nabla r_1\|_{L^2}^2. \end{aligned}$$

Remark. Here one could get a stronger control by keeping track of the loss in the last step (which is proportional to $\|r_1\|_2^2$), but it is not needed.

We just need to estimate these two coefficients: let $Y \in \{1, a \cdot \omega\}$ for $a \in \mathbb{R}^{n+1}$, then

$$\left| \int_{\mathbb{S}^n} Y r_1 d\sigma \right| = \left| \int_{\mathbb{S}^n} Y (r_2 + r_3) d\sigma \right| \leq \|Y\|_4 |\{r_1 \geq \varepsilon_1\}|^{\frac{1}{4}} \|r_2 + r_3\|_2$$

by Hölder inequality ($\frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$), and we can easily verify that

$$\begin{aligned} \|r_2 + r_3\|_2^2 &\leq \|r\|_q \|r_2 + r_3\|_2 \leq \delta \sqrt{2} \|r_2^2 + r_3^2\|_1 \\ |\{r_1 \geq \varepsilon_1\}| &\leq \frac{1}{\varepsilon_1^2} \|r_1\|_2^2 \\ \|1\|_4 &= 1 \quad \|a \cdot \omega\|_4^4 = \frac{3|a|^4}{(n+1)(n+3)} \leq \frac{3|a|^4}{(n+1)^2}. \end{aligned}$$

Putting all the pieces together gives

$$\left| \int_{\mathbb{S}^n} r_1 d\sigma \right|^2 \leq \frac{\delta \sqrt{2}}{\varepsilon_1} \|r_1\|_2 \|r_2^2 + r_3^2\|_1^{\frac{1}{2}} \quad (n+1) \left| \int_{\mathbb{S}^n} \omega r_1 d\sigma \right|^2 \leq \frac{\delta \sqrt{6}}{\varepsilon_1} \|r_1\|_2 \|r_2^2 + r_3^2\|_1^{\frac{1}{2}},$$

so that

$$\begin{aligned} I_1 &\geq (2(n+1)(1 - \varepsilon_0 \theta) - n(1 + \varepsilon_0 + \varepsilon_1)) \|r_1\|_2^2 \\ &\quad - 2(n+1)(1 - \varepsilon_0 \theta) \frac{\delta(\sqrt{2} + \sqrt{6})}{\varepsilon_1} \|r_1\|_2 \|r_2^2 + r_3^2\|_1^{\frac{1}{2}} + n\sigma_0 \|r_2^2 + r_3^2\|_1, \end{aligned}$$

which, after defining $x = \|r_1\|_2$ and $y = \|r_2^2 + r_3^2\|_1^{\frac{1}{2}}$, can be simplified to

$$\frac{1}{n} I_1 \geq (1 - 3\varepsilon_0 - \varepsilon_1) x^2 - 10 \frac{\delta}{\varepsilon_1} x y + \sigma_0 y^2.$$

Finally, as long as $3\varepsilon_0 + \varepsilon_1 \leq \frac{1}{2}$, a sufficient condition for having negative discriminant is

$$50 \frac{\delta^2}{\varepsilon_1^2} \leq \sigma_0. \quad (4.3)$$

4.2.2 Second term

In order to deal with this piece, we will need the following estimate.

Theorem 4.2.1 (Duoandikoetxea (see [24])). *For any k -th degree spherical harmonic*

$$\|Y_k\|_{L^p(\mathbb{S}^n)} \leq (p-1)^{\frac{k}{2}} \|Y_k\|_{L^2(\mathbb{S}^n)} .$$

Given any k -th degree spherical harmonic Y , let us bound as before

$$\left| \int_{\mathbb{S}^n} Y r_2 d\sigma \right| \leq \|Y\|_4 |\{r \geq \varepsilon_1\}|^{\frac{1}{4}} \|r_2\|_2 ,$$

but this time we will put together the following pieces

$$|\{r \geq \varepsilon_1\}| \leq \frac{1}{\varepsilon_1^q} \|r\|_q^q \leq \frac{\delta^q}{\varepsilon_1^q} \quad \|Y\|_4 \leq 3^{\frac{k}{2}} .$$

Indeed, if π_k is the projection on the k -th order spherical harmonics we get

$$\|\pi_k r_2\|_2 \leq \frac{3^{\frac{k}{2}} \delta^{\frac{q}{4}}}{\varepsilon_1^{\frac{q}{4}}} \|r_2\|_2 ,$$

and in turn, letting $\Pi_K = \sum_{k=0}^{K-1} \pi_k$,

$$\|\Pi_K r_2\|_2 \leq \frac{3^{\frac{K}{2}} \delta^{\frac{q}{4}}}{\varepsilon_1^{\frac{q}{4}}} \|r_2\|_2 .$$

Finally

$$\|\nabla r_2\|_2^2 \geq \|\nabla(1 - \Pi_K) r_2\|_2^2 \geq K(K+n-1) \|(1 - \Pi_K) r_2\|_2^2 \geq Kn \left(1 - \frac{3^K \delta^{\frac{q}{2}}}{\varepsilon_1^{\frac{q}{2}}} \right) \|r_2\|_2^2 ,$$

and one can first choose any (universally bounded)

$$K \geq 2 \frac{1 + \varepsilon_0 + \sigma_0 + C_{\varepsilon_1, \varepsilon_2}}{1 - \varepsilon_0} , \tag{4.4}$$

where RHS is at least twice the constant appearing in I_2 , and then δ universally small to make the last coefficient greater than $\frac{1}{2}$.

Remark. Note that, except for the condition $2\varepsilon_1 < \varepsilon_2$, all the other constraints are clearly satisfied by just imposing $\max\{\varepsilon_0, \varepsilon_1, \varepsilon_2, \sigma_0\} \leq C$ for an appropriate $C > 0$ universal and $\delta \leq C(\varepsilon_0, \varepsilon_1, \varepsilon_2, \sigma_0)$.

4.3 Local to Global

Now let us prove the existence of a $C(n) \leq Cn$ for 4.1 for all $f \in \dot{H}^1(\mathbb{S}^n)$ such that $f \geq 0$ (implicit in all the statements). First let us see how to transfer the local estimate to $\mathbb{R}^n$.

Lemma 4.3.1. *If $d(f, \mathcal{M})^2 \leq \delta_1 \|\nabla f\|_2^2$ with $\delta_1 = \frac{\delta}{1+\delta}$, then $\mathcal{E}(f) \geq \varepsilon_0 \theta$.*

Proof. By conformal invariance we can assume that $d(f, \mathcal{M})$ is attained at $g_0 = \frac{1}{(1+|x|^2)^{\frac{n-2}{2}}}$, which, up to rescaling, corresponds to the constant 1 when we reading the problem on $\mathbb{S}^n$. Define $r = F - 1$, then the hypothesis reads as

$$\|\nabla r\|_2^2 + A \|r\|_2^2 \leq \frac{\delta}{1+\delta} (\|\nabla r\|_2^2 + A \|1+r\|_2^2) = \frac{\delta}{1+\delta} (A + \|\nabla r\|_2^2 + A \|r\|_2^2),$$

or equivalently

$$\|\nabla r\|_2^2 + \|r\|_2^2 \leq \delta A,$$

which implies $\|r\|_{2^*} \leq \delta$ by the sharp Sobolev inequality. $\square$

From now on, assume instead that $d(f, \mathcal{M})^2 > \delta_1 \|\nabla f\|_2^2$. Consider the sequence $f_k = (\mathcal{R}U)^k f$ introduced in the appendix, and recall that $f \mapsto \mathcal{R}f$ can be seen as a countable composition of Steiner symmetrizations. Therefore, the elements of the sequence f_k can be connected by a curve $t \mapsto f_t \in L^{2^*}(\mathbb{S}^n)$ satisfying the same properties of the continuous Steiner symmetrization.

We need a couple of lemmas regarding this sequence.

Lemma 4.3.2. *For every positive $f \in \dot{H}^1(\mathbb{R}^n)$*

$$d(f, \mathcal{M})^2 = \|\nabla f\|_2^2 - S_n \sup_{\mathcal{M}_1} \left(\int_{\mathbb{S}^n} f g^{2^*-1} d\sigma \right)^2. \quad (4.5)$$

Proof. Recall that every $g \in \mathcal{M}_1$ satisfies the Sobolev equation

$$-\Delta g = S_n \frac{g^{2^*-1}}{\|g\|_{2^*}^{2^*-2}} = S_n g^{2^*-1} = S_n g^{2^*-1},$$

and $\|\nabla g\|_2^2 = S_n \|g\|_{2^*}^2 = S_n$, therefore, for any $c \in \mathbb{R}$,

$$\|\nabla f - c \nabla g\|_2^2 = \|\nabla f\|_2^2 - 2c \int \nabla f \cdot \nabla g + c^2 \|\nabla g\|_2^2 = \|\nabla f\|_2^2 + 2c S_n \int f g^{2^*-1} + S_n c^2.$$

The minimum in $c \in \mathbb{R}$ is precisely

$$\|\nabla f\|_2^2 - S_n \left(\int_{\mathbb{S}^n} f g^{2^*-1} d\sigma \right)^2,$$

from which the identity follows. $\square$

Lemma 4.3.3. *The sequence $d(f_k, \mathcal{M})^2$ is strictly decreasing and it converges to $\lim_k \|\nabla f_k\|_2^2 - S_n \|f\|_{2^*}^2$.*

Proof. Applying U does not affect the values involved, so we just need to discuss the application of $\mathcal{R}$. Look at the identity 4.5:

- The first term is decreasing by the Pólya-Szegő inequality.
- Recall from the appendix that $f_k \rightarrow cg_0$ in L^{2^*} for $c > 0$ such that $\|f\|_{2^*} = \|cg_0\|_{2^*}$, therefore the second term coincides with $S_n \|f\|_{2^*}^2$ by Hölder inequality and $g_0 \in \mathcal{M}$. Moreover it is strictly increasing due to the properties of $\mathcal{R}$ and because the functions in $\mathcal{M}$ are radially decreasing up to translation.

□

In order to conclude, let us distinguish two cases:

- $d(f_k, \mathcal{M}) \geq \delta_1 \|\nabla f_k\|_2$ for all n : since $d(f, \mathcal{M}) \leq \|\nabla f\|_2$, we have

$$\mathcal{E}(f) \geq \frac{\|\nabla f\|_2^2 - S_n \|f\|_{2^*}^2}{\|\nabla f\|_2^2} \geq \frac{\lim_k \|\nabla f_k\|_2^2 - S_n \|f\|_{2^*}^2}{\lim_k \|\nabla f_k\|_2^2} \geq \delta_1.$$

- Otherwise let k be an index such that

$$d(f_k, \mathcal{M})^2 \geq \delta_1 \|\nabla f_k\|_2^2 \quad d(f_{k+1}, \mathcal{M})^2 < \delta_1 \|\nabla f_{k+1}\|_2^2.$$

Then let us consider the usual curve $[0, 1] \ni t \mapsto g_t \in L^{2^*}(\mathbb{R}^n)$ from Uf_k to f_{k+1} made of continuous Steiner symmetrizations.

Remark. The idea is the following: if there was an s such that $d(g_s, \mathcal{M})^2 = \delta_1 \|g_s\|_2^2$, then we could perform the following computation

$$\mathcal{E}(f) \geq \frac{\|\nabla f\|_2^2 - S_n \|f\|_{2^*}^2}{\|\nabla f\|_2^2} \geq \frac{\|\nabla g_s\|_2^2 - S_n \|f\|_{2^*}^2}{\|\nabla g_s\|_2^2} = \delta_1 \mathcal{E}(g_s) \geq \delta_1 \varepsilon_0 \theta$$

The main issue is that none of these terms are continuous in t , and we can only rely on the right continuity of $\|\nabla g_s\|_2$ (see the appendix).

Indeed, let

$$\tau = \inf \{s \in [0, 1] : d(g_s, \mathcal{M})^2 < \delta_1 \|\nabla g_s\|_2^2\}.$$

Then by identity 4.5 and right continuity we get $d(g_\tau, \mathcal{M})^2 \leq \delta_1 \|\nabla g_\tau\|_2^2$. Finally, similarly to before, given $\ell = \lim_{s \rightarrow \tau^-} \|\nabla g_s\|_2^2 \geq \|\nabla g_\tau\|_2^2$ and assuming $g_\tau \notin \mathcal{M}$, we have

$$\mathcal{E}(f) \geq \delta_1 \frac{\ell - S_n \|g_\tau\|_{2^*}^2}{\lim_{s \rightarrow \tau^-} d(g_s, \mathcal{M})^2} = \delta_1 \frac{\ell - S_n \|g_\tau\|_{2^*}^2}{\ell - S_n \sup_{g \in \mathcal{M}_1} \langle g_\tau, g^{2^*-1} \rangle^2} \geq \delta_1 \mathcal{E}(g_\tau) \geq \delta_1 \varepsilon_0 \theta,$$

where the monotonicity in the second to last step holds because $\|g_\tau\|_{2^*} \geq \sup_{g \in \mathcal{M}_1} \langle g_\tau, g^{2^*-1} \rangle$ by Hölder inequality. If $g_\tau \in \mathcal{M}$ then the third term is precisely δ_1 (numerator and denominator are equal).

4.4 General Sign

Finally, let $f \in \dot{H}^1(\mathbb{R}^n)$ be any function with $\|f\|_{2^*} = 1$ and $m = \|f_-\|_{2^*}^{2^*} \in [0, \frac{1}{2}]$ without loss of generality. The thesis will follow from a standard concavity argument, analogous to the isoperimetric case.

Indeed, for any function h , let us define $D(h) = \|\nabla h\|_2^2 - S_n \|h\|_{2^*}^2$. Setting $\alpha_n = \frac{2}{2^*} = 1 - \frac{2}{n}$, we introduce the function

$$\psi(t) = t^{\alpha_n} + (1-t)^{\alpha_n} - 1,$$

which satisfies $\psi(m) \geq (2 - 2^{\alpha_n})m^{\alpha_n}$ analogously to 2.6. In conclusion

$$\begin{aligned} D(f) &= D(f_+) + D(f_-) + S_n \psi(m) \\ &\geq D(f_+) + (2 - 2^{\alpha_n})(D(f_-) + S_n m^{\alpha_n}) \\ &= D(f_+) + (2 - 2^{\alpha_n}) \|\nabla f_-\|_2^2 \\ &\geq \min\{\delta_1 \varepsilon_0 \theta, 2 - 2^{\alpha_n}\} (d(f_+, \mathcal{M})^2 + \|\nabla f_-\|_2^2) \\ &\geq \frac{1}{2} \min\{\delta_1 \varepsilon_0 \theta, 2 - 2^{\alpha_n}\} d(f, \mathcal{M})^2. \end{aligned}$$

Chapter 5

The Current Attempt

5.1 Refining Fuglede

For notation purposes, let us phrase the problem in $\mathbb{S}^n \subset \mathbb{R}^{n+1}$ and normalize the surface measure σ of $\mathbb{S}^n$.

An algebraic observation that makes the perimeter functional for a graphical set more manageable is the following: if E is described by u , then after defining

$$r = (1 + u)^{\frac{n}{2}} - 1,$$

we have

$$\frac{P(E)}{(n+1)\omega_{n+1}} = \int_{\mathbb{S}^n} (1+r)^2 \sqrt{1 + \frac{4}{n^2} \frac{|\nabla r|^2}{(1+r)^2}} d\sigma$$

and

$$\frac{|E|}{\omega_{n+1}} = \int_{\mathbb{S}^n} (1+r)^q d\sigma$$

for $q = 2\frac{n+1}{n} = 2 + \frac{2}{n}$ (and $\theta = \frac{2}{n}$). Then the sharp isoperimetric inequality reads as

$$\frac{\delta(E)}{n\omega_n} = \int (1+r)^2 \sqrt{1 + \frac{4|\nabla r|^2}{n^2(1+r)^2}} d\sigma - \left(\int (1+r)^q d\sigma \right)^{\frac{2}{q}} \geq 0$$

for every $r \in H^1(\mathbb{S}^n)$ such that $r \geq -1$ or equivalently, setting $h = 1 + r$,

$$\int h^2 \sqrt{1 + \frac{4|\nabla h|^2}{n^2 h^2}} d\sigma \geq \|h\|_q^2 \quad (5.1)$$

for any $h \geq 0$. Note that it also holds for every $h \in H^1(\mathbb{S}^n)$ (by plugging $|h|$). Moreover

$$\alpha(E) \leq \frac{|E\Delta B|}{|E|} \leq \frac{\omega_{n+1}}{|E|} \int_{\mathbb{S}^n} |(1+r)^q - 1| d\sigma \lesssim \int_{\mathbb{S}^n} |(1+r)^q - 1| d\sigma,$$

where the last step is possible whenever we assume that $|E| \asymp \omega_{n+1}$, which will be the case for us (note that we are not assuming $|E| = \omega_{n+1}$).

5.1.1 With Lipschitz Bound

In this framework, we will discuss a possible approach to look for a asymptotically sharp quantitative isoperimetric inequality for sets satisfying $\|\nabla u\|_\infty \leq \varepsilon$ universal (stronger than Fuglede's result). First note that

$$\frac{\nabla r}{1+r} = \frac{n}{2} \frac{\nabla u}{1+u}$$

and consequently $\left\| \frac{\nabla r}{1+r} \right\|_\infty \lesssim \varepsilon n$.

As previously mentioned, we want to manage all sets with a universally small asymmetry index. In this framework, however, the natural quantity to bound is directly $\|r\|_q$ rather than $\alpha(E)$. This is not a major issue: indeed, if $\|r\|_q \leq 1$ we have

$$\alpha(E) \lesssim \|r\|_1 + \|r\|_q^q \lesssim \|r\|_q ,$$

and on the other end $x^q \lesssim |(1+x)^q - 1|$ for all $x \geq -1$, therefore

$$\|r\|_q^q \lesssim \alpha(E) .$$

Hence, trying to estimate $\|r\|_q$ is reasonable, not only because it would imply an analogous estimate for $\alpha(E)$, but also because a universal asymmetry bound $\alpha(E) \leq \varepsilon$ corresponds to a bound for $\|r\|_q$ which is still universal.

Moreover, imposing that E has volume ω_n and barycenter in the origin can be stated in terms of r , but it is not the cleanest condition. For this reason, for the moment, we only considered hypotheses which directly regard r itself. While it is possible to translate the smallness conditions from one formulation to the other, we will stick to this setting to avoid unnecessary technical noise.

Let us show a first naive approach to this idea.

Theorem 5.1.1. *There exists a universal constant $\varepsilon > 0$ such that, if $\|\nabla u\|_\infty \leq \frac{\varepsilon}{\sqrt{n}}$, $\|r\|_q \leq \varepsilon$ and*

$$\int_{\mathbb{S}^n} r \, d\sigma = 0 \quad \int_{\mathbb{S}^n} r x \, d\sigma(x) = 0 ,$$

then

$$C\theta \|r\|_q^2 \leq \delta(E)$$

for some universal constant $C > 0$.

Proof. To begin,

$$\|1+r\|_2^2 + \left(1 - C\frac{\varepsilon}{n}\right) \frac{2}{n^2} \|\nabla r\|_2^2 \leq \frac{P(E)}{(n+1)\omega_{n+1}} \leq \|1+r\|_2^2 + \left(1 + C\frac{\varepsilon}{n}\right) \frac{2}{n^2} \|\nabla r\|_2^2 . \quad (5.2)$$

The idea is precisely to apply the left side of 5.2 and then follow the proof of the Sobolev case by Dolbeault et al.: divide $r = r_1 + r_2 + r_3$ in the same way, and redefine $A = \frac{n^2}{2}$ and $\theta = \frac{2}{n}$, then following the same computations one can check that $\frac{n^2}{2}\delta(E)$ is greater than

$$\varepsilon_0\theta \left(\|\nabla r\|_2^2 + \frac{n^2}{2} \|r\|_2^2 \right) + I_1 + I_2 + I_3 ,$$

where

$$\begin{aligned} I_1 &= (1 - 2C\varepsilon\theta - \varepsilon_0\theta) \|\nabla r_1\|_2^2 - n(1 + \varepsilon_0 + \varepsilon_1) \|r_1\|_2^2 + n\sigma_0(\|r_2\|_2^2 + \|r_3\|_2^2) \\ I_2 &= (1 - 2C\varepsilon\theta - \varepsilon_0\theta) \|\nabla r_2\|_2^2 - n(1 + \varepsilon_0 + \sigma_0 + C_{\varepsilon_1, \varepsilon_2}) \|r_2\|_2^2 \\ I_3 &= (1 - 2C\varepsilon\theta - \varepsilon_0\theta) \|\nabla r_3\|_2^2 + A(1 - (\sigma_0 + \varepsilon_0)\theta) \|r_3\|_2^2 - \frac{2}{q}A(1 + \varepsilon_2\theta) \|r_3\|_q^q . \end{aligned}$$

Since we have a new contribution proportional to θ , the computations of the original proof can be carried out in the same exact way for a suitable $\varepsilon > 0$ universally small enough.

Finally the isoperimetric inequality 5.1, together with the right hand side of 5.2, gives

$$\|r\|_2^2 + \frac{n^2}{2} \|\nabla r\|_2^2 \gtrsim \frac{n^2}{2} \|r\|_q^2 .$$

□

We would like this approach to work also for the weaker condition $\|\nabla u\|_\infty \leq \varepsilon$. However, this would mean that the estimate 5.2 holds with $C\varepsilon$ instead of $C\frac{\varepsilon}{n}$, and consequently we would carry a factor $C\varepsilon$ inside the terms I_j . Unfortunately, we note the following issues:

- I_1 and I_2 can be bounded in the same way, because in both cases one can rely on harmonics with big enough degree. For example, intuitively $I_1 \simeq \|\nabla r_1\|_2^2 - (1 + C\varepsilon)n \|r_1\|_2^2$, but we can ensure that (up to small terms) $\|\nabla r_1\|_2^2 \geq 2n(1 - C\varepsilon) \|r_1\|_2^2$.
- Instead I_3 is problematic because we would like to reduce to the isoperimetric inequality via the right side of 5.2, but the contribution of $C\varepsilon$ cannot be absorbed by terms with a factor $\theta = \frac{2}{n}$.

In particular, this decomposition does not seem capable of proving the bound under our weaker hypotheses.

Note that the condition $\|\nabla u\|_\infty \leq \frac{\varepsilon}{\sqrt{n}}$ is not directly comparable with Fuglede's hypotheses $\|u\|_\infty \leq \frac{\varepsilon}{n}$ and $\|\nabla u\|_\infty \leq \varepsilon$, therefore this is a new, but weak, result.

5.1.2 With Volume Bound Only

Our hope is to somehow further drop the Lipschitz bound in the previous attempt, therefore one would like to nicely bound the perimeter functional without relying on a Taylor expansion, but only on monotonicities.

In particular, following the previous computations, it is clear that the loss in 5.2 is not an intrinsic obstacle of this problem, but instead a symptom that this is not the correct point of view.

At the moment, this approach remains a work in progress.

5.2 From Local to Global

Let us briefly discuss the main obstacles we face:

- We can restrict to $\alpha(E) \leq \varepsilon$ for any fixed universal $\varepsilon > 0$: indeed, let $E_n = S_n E_{n-1}$ be a sequence of Steiner symmetrizations converging to a ball B , then, analogously to the Sobolev case, they define a curve $t \mapsto E_t$ which is continuous in $L^1(\mathbb{R}^n)$, such that $|E_t|$ is constant, $P(E_t)$ is decreasing and $E_t \rightarrow B$. Then there exists an index s such that $\alpha(E_s) = \varepsilon$, therefore

$$\frac{\alpha(E)}{\sqrt{\delta(E)}} \leq \frac{2}{\varepsilon} \frac{\alpha(E_s)}{\sqrt{\delta(E_s)}}.$$

A significant limitation of this approach is that it only restricts to sets which are round in measure, lacking any additional regularity properties.

The Sobolev case was significantly different because it was possible to perform the local expansion for any sufficiently close function.

- The selection principle by Cicalese and Leonardi is not the right tool for us, for a few evident reasons:
 - One needs to restrict to uniformly bounded sets, a step which already adds a large dimensional constant.
 - The result by Tamanini is reminiscent of other well known regularity results, like Allard's theorem for stationary varifolds and Schauder's regularity for uniformly elliptic PDEs. In all these cases, the proof strategy is the same, and it is clear that any quantitative estimate is either impossible or at most extremely weak.
 - The proof relies intrinsically on a contradiction argument. Indeed, the selection principle makes us lose track of any original convergence rate, given that $F_j \rightarrow B$ is obtained with the standard subsequence argument.
- An alternative strategy involves applying a star-shaped symmetrization to E , potentially developing a continuous version of this transformation.

A natural first attempt is to fix an optimal ball and perform a one dimensional (continuous) Steiner symmetrization on every line passing through the center. More precisely

$$E^* = \bigcup_{\nu \in \mathbb{S}^{n-1}} \left[-\frac{\mathcal{H}^1(E \cap \mathbb{R}\nu)}{2}, \frac{\mathcal{H}^1(E \cap \mathbb{R}\nu)}{2} \right] \nu,$$

which is also graphical in Fuglede's sense. It is in fact well known that this process decreases the perimeter (and one expects the continuous version to do the same). However it can also drastically decrease the volume, since the formula in spherical coordinates $\mathbb{R} \times \mathbb{S}^{n-1}$ involves the rapidly increasing factor t^{n-1} .

Consequently, one could hope to use the volume-fixing version of this transformation, namely

$$E_V^* = \bigcup_{\nu \in \mathbb{S}^{n-1}} \left[-\frac{\ell(E, \nu)}{2}, \frac{\ell(E, \nu)}{2} \right] \nu,$$

where

$$\frac{1}{n} \ell(E, \nu)^n = \int_{E_\nu} t^{n-1} dt$$

with $E_\nu = \{t \in \mathbb{R} : t\nu \in E\}$. However, Kawohl found explicit examples of sets (and functions) satisfying $P(E) > P(E_V^*)$ (see [34]). In these contexts, the correct inequality to expect would concern an appropriate weighted perimeter. Many instances of this phenomenon have been found in more recent years.

Appendix A

(Continuous) Symmetrizations

Symmetrizations have been revolutionary tools in the study of shape optimization problems. In this section we present some of their properties and, most importantly, we define the continuous Steiner symmetrization, first introduced by Brock (see [12], [13]).

A standard reference for classical symmetrizations is [36].

Theorem A.0.1 (Brock). *Let H be a hyperplane in $\mathbb{R}^n$ and $D = \mathbb{R}_+ \cup \{+\infty\}$. There exists a family of operators $\{M_t\}_{t \in D}$ on $L_+^p(\mathbb{R}^n)$ for any $1 \leq p < \infty$ such that:*

1. $M_0 = \text{Id}$ and $M_\infty = S_H$, where S_H is the Steiner symmetrization with respect to H .
2. The operators satisfy $M_t \circ M_s = M_{t+s}$ for all $t, s \in D$.
3. $M_t(f)$ is equimeasurable with f for all $t \in D$.
4. For any $f, g \in L_+^p(\mathbb{R}^n)$ we have

$$\|M_t(f) - M_t(g)\|_{L^p(\mathbb{R}^n)} \leq \|f - g\|_{L^p(\mathbb{R}^n)}$$

for all $t \in D$.

5. For any $f \in L_+^p(\mathbb{R}^n)$ and $g \in L_+^q(\mathbb{R}^n)$ with $\frac{1}{p} + \frac{1}{q} = 1$, we have

$$\int_{\mathbb{R}^n} fg \, dx \leq \int_{\mathbb{R}^n} M_t(f) M_t(g) \, dx$$

for all $t \in D$.

6. If $f \in \dot{W}^{1,p}(\mathbb{R}^n)$ and $f \geq 0$, then $M_t(f) \in \dot{W}^{1,p}(\mathbb{R}^n)$, and the curve $t \mapsto \|\nabla M_t(f)\|_{L^p(\mathbb{R}^n)}$ is non-increasing.

7. The curve $t \mapsto M_t(f)$ is continuous in $L^p(\mathbb{R}^n)$.
8. If $f \in \dot{H}^1(\mathbb{R}^n)$ and $f \geq 0$, the curve $t \mapsto \|\nabla M_t(f)\|_{L^2(\mathbb{R}^n)}$ is right-continuous (see [23]).

The construction is initially defined for positive functions in $L^p(\mathbb{R})$. For a general positive function in $L^p(\mathbb{R}^n)$, we then apply it to every line orthogonal to H . The idea is the following:

- If $f = \chi_E$ with $E = \bigsqcup_{k=1}^j I_k$ for some open intervals I_k , we translate their centers towards the origin: more precisely, if $I = b + (-a, a)$ we consider $t \mapsto e^{-t}b + (-a, a)$, which is exponential for composition purposes. The first time two of them touch, say t_0 , we will have a new decomposition $E_{t_0} = \bigsqcup_{k=1}^{j'} I'_k$ and we restart the flow analogously (clearly with respect to the new centers).
- If $f = \sum_{k=1}^j c_k \chi_{I_k}$ is a positive simple function, the level sets are finite unions of open intervals, hence we can define $M_t(f)$ via the layer-cake formula:

$$M_t(f) = \int_0^\infty \chi_{M_t(\{f>t\})} dt.$$

- Finally, one can easily check that $\|M_t(f) - M_t(g)\|_p \leq \|f - g\|_p$ for all simple functions f, g and $1 \leq p < \infty$, therefore we can define M_t on the whole $L^p_+(\mathbb{R})$ by density.

Theorem A.0.2 (Brascamp, Lieb, Luttinger (see [11])). *Given any function $f \in L^p(\mathbb{R}^n)$ there exists a sequence of Steiner symmetrizations S_k (of hyperplanes passing through the origin) such that $f_k = S_k \dots S_1 f \rightarrow \mathcal{R}f$ in $L^p(\mathbb{R}^n)$.*

Proof. Inductively choose the k -th H_k hyperplane in such a way that

$$\|S_k f_{k-1} - \mathcal{R}f\|_p \leq \inf_H \|S_H f_{k-1} - \mathcal{R}f\|_p + \frac{1}{k}.$$

Moreover observe that $\inf_H \|S_H f - \mathcal{R}f\|_p < \|f - \mathcal{R}f\|_p$ whenever $f \neq \mathcal{R}f$. It is well known that Steiner symmetrizations do not increase the L^p modulus of continuity, so that $\|\tau_h f_k - f_k\|_p \leq \|\tau_h f - f\|_p$ for all $h \in \mathbb{R}^n$ and in particular the sequence is equicontinuous. Moreover, one can also prove that f_k is equitight, therefore (f_k) is precompact in $L^p(\mathbb{R}^n)$ by the Riesz-Fréchet-Kolmogorov theorem.

Let $f_{k_j} \rightarrow g$ be a subsequence converging in $L^p(\mathbb{R}^n)$. If by contradiction $g \neq \mathcal{R}f$ (also note that $\mathcal{R}g = \mathcal{R}f$), then there exists a symmetrization S such that $\|Sg - \mathcal{R}f\|_p = \|g - \mathcal{R}f\|_p - \varepsilon$ for some $\varepsilon > 0$. Because $f_{k_j} \rightarrow g$ strongly, for j large enough we have

$$\|f_{k_j} - \mathcal{R}f\|_p - \frac{1}{k_j} \leq \|S f_{k_j-1} - \mathcal{R}f\|_p \leq \|f_{k_j-1} - \mathcal{R}f\|_p - \frac{\varepsilon}{2},$$

and in turn also $\|f_{k_j} - \mathcal{R}f\|_p \leq \|f_{k_j-1} - \mathcal{R}f\|_p - \frac{\varepsilon}{4}$, which is impossible. Finally, since every convergent subsequence must converge to $\mathcal{R}f$ and the sequence is precompact, we have $f_k \rightarrow \mathcal{R}f$ in $L^p(\mathbb{R}^n)$. $\square$

Let $\mathcal{R}$ be the radial rearrangement on $\mathbb{R}^n$ and U the conformal transformation on $\mathbb{R}^n$ such that

$$Uf(x) = \left(\frac{2}{|x - e_n|^2} \right)^{\frac{d-2}{2}} f \left(\frac{1}{|x - e_n|^2} (x_1, \dots, x_{n-1}, |x|^2 - 1) \right),$$

which, when read on the sphere $\mathbb{S}^n$ via stereographic projection, corresponds to a rotation of the last two components:

$$(UF)(\omega_1, \dots, \omega_n, \omega_{n+1}) = F(\omega_1, \dots, \omega_{n+1}, -\omega_n).$$

It is easy to check that U is an isometry of both $L^{2^*}(\mathbb{R}^n)$ and $\dot{H}^1(\mathbb{R}^n)$, and in turn it preserves the ratio $\frac{\|f\|_{2^*}}{\|\nabla f\|_2}$, whereas $\mathcal{R}$ is an isometry in any $L^p(\mathbb{R}^n)$ and satisfies the Pólya-Szegő inequality

$$\|\nabla(\mathcal{R}f)\|_{L^2(\mathbb{R}^n)} \leq \|\nabla f\|_{L^2(\mathbb{R}^n)}.$$

For example, it easily follows from applying the sharp isoperimetric inequality to the coarea formula for f . From these properties we deduce that the ratio $\frac{\|f\|_{2^*}}{\|\nabla f\|_2}$ is non-decreasing when applying $\mathcal{R}$.

Theorem A.0.3 (Carlen, Loss (see [16])). *Given any non-negative function $f \in \dot{H}^1(\mathbb{R}^n)$, the sequence $f_k = (\mathcal{R}U)^k f$ converges in $L^{2^*}(\mathbb{R}^n)$ to the Aubin-Talenti bubble*

$$g = \frac{c}{(1 + |x|^2)^{\frac{n-2}{2}}}$$

with $c > 0$ such that $\|g\|_{L^{2^*}} = \|f\|_{L^{2^*}}$.

Proof. By the non-expansivity of $\mathcal{R}$, it is sufficient to prove the theorem for a dense subset of $L^{2^*}(\mathbb{R}^n)$, so we can assume that f is bounded with bounded support. Now note that both $\mathcal{R}$ and U are monotone operators, therefore if $f \leq Cg$ for some $C > 0$, then we also have $f_k \leq Cg$ and consequently f_k is dominated by a function in L^{2^*} . Moreover the functions f_k are all radially decreasing, therefore there exists a subsequence k_j such that $f_{k_j} \rightarrow h$ a.e. for some h by Helly's selection theorem, and also in L^{2^*} by dominated convergence (we do not know that $\mathcal{R}Uh = h$ yet because we passed to a subsequence). Finally note that $\|f_{k_{j+1}} - g\|_{2^*} \leq \|\mathcal{R}Uf_{k_j} - g\|_{2^*}$ and in turn

$$\|h - g\|_{2^*} \leq \|\mathcal{R}Uh - g\|_{2^*} \leq \|Uh - g\|_{2^*} = \|h - g\|_{2^*},$$

which means that Uh is radially decreasing, but h is also radially decreasing and this can only happen if $h = g$. $\square$

Remark (Sharp Sobolev inequality). Since the ratio $\frac{\|f_k\|_{2^*}}{\|\nabla f_k\|_2}$ is increasing during this procedure and $\lim_k \frac{\|f_k\|_{2^*}}{\|\nabla f_k\|_2} \leq S_n$ (the ratio of any Aubin-Talenti bubble) by weak lower semi-continuity, it means that the optimal constant for the Sobolev inequality is precisely S_n .

Appendix B

Finite Perimeter Sets

A standard reference for the material presented in this appendix is the monograph by Ambrosio, Fusco, and Pallara [3].

We begin by introducing the space of functions of bounded variation and the decomposition of their distributional gradient.

Definition B.0.1. *Let $\Omega \subset \mathbb{R}^n$ be an open set. A function $u \in L^1(\Omega)$ is said to be of bounded variation, denoted by $u \in BV(\Omega)$, if its distributional gradient Du is a finite vector-valued Radon measure on Ω .*

A set $E \subset \mathbb{R}^n$ is a set of finite perimeter in Ω if its characteristic function χ_E belongs to $BV(\Omega)$. The perimeter of E in Ω is defined as $P(E, \Omega) = |D\chi_E|(\Omega)$.

Theorem B.0.1. *Let $u \in BV(\mathbb{R}^n)$ and let*

$$Du = D^a u + D^s u = \nabla u \mathcal{L}^n + D^s u$$

be the Lebesgue decomposition of Du with respect to $\mathcal{L}^n$, then the singular measure $D^s u$ can be uniquely split into a Cantor part $D^c u$ and a jump part $D^j u$, satisfying the following properties:

1. *$D^j u$ is concentrated on the approximate jump set S_u and is given by*

$$D^j u = (u^+ - u^-) \nu_u \mathcal{H}^{n-1} \llcorner S_u,$$

where u^+ and u^- are the upper and lower approximate limits of u along the direction of the measure-theoretic normal ν_u of S_u .

2. *$D^c u$ is singular with respect to $\mathcal{L}^n$ and moreover $|D^c u|(A) = 0$ for any Borel set A with $\mathcal{H}^{n-1}(A) < \infty$.*

Definition B.0.2. *Let $E \subset \mathbb{R}^n$ be a Lebesgue measurable set. For any $t \in [0, 1]$, the set of points of density t is defined as*

$$E^{(t)} = \left\{ x \in \mathbb{R}^n : \lim_{\rho \rightarrow 0} \frac{\mathcal{L}^n(E \cap B_\rho(x))}{\mathcal{L}^n(B_\rho(x))} = t \right\}.$$

The essential boundary of E is defined as

$$\partial_* E = \mathbb{R}^n \setminus \left(E^{(0)} \cup E^{(1)} \right).$$

Definition B.0.3. Let E be a set of finite perimeter in $\mathbb{R}^n$. The reduced boundary of E , denoted by $\partial^* E$, is the set of points $x \in \text{supp}(|D\chi_E|)$ such that the limit

$$\nu_E(x) = - \lim_{\rho \rightarrow 0} \frac{D\chi_E(B_\rho(x))}{|D\chi_E|(B_\rho(x))}$$

exists and satisfies $|\nu_E(x)| = 1$.

We now report the structure theorem for the reduced boundary due to De Giorgi.

Theorem B.0.2. Let $E \subset \mathbb{R}^n$ be a set of finite perimeter, then:

1. $D\chi_E$ is concentrated on the reduced boundary $\partial^* E$ and satisfies

$$D\chi_E = -\nu_E \mathcal{H}^{n-1} \llcorner \partial^* E.$$

In particular $P(E) = \mathcal{H}^{n-1}(\partial^* E)$.

2. The reduced boundary $\partial^* E$ is countably $(n-1)$ -rectifiable, meaning that

$$\partial^* E = N \cup \bigcup_{h=1}^{\infty} K_h,$$

where $\mathcal{H}^{n-1}(N) = 0$ and each K_h is a compact subset of a C^1 hypersurface.

3. For every $x \in \partial^* E$, the dilated sets $E_{x,\rho} = \frac{E-x}{\rho}$ converge in $L^1_{\text{loc}}(\mathbb{R}^n)$ as $\rho \rightarrow 0$ to the inner tangent halfspace $H_{\nu_E(x)}$, defined by

$$H_{\nu_E(x)} = \{y \in \mathbb{R}^n : y \cdot \nu_E(x) \leq 0\}.$$

Theorem B.0.3. Let $E \subset \mathbb{R}^n$ be a set of finite perimeter. Then

$$\partial^* E \subset E^{(1/2)} \subset \partial_* E$$

and

$$\mathcal{H}^{n-1}(\partial_* E \setminus \partial^* E) = 0.$$

Next, we recall the behavior of functions of bounded variation under traces, compositions, and product operations.

Theorem B.0.4. *If $R \subset \mathbb{R}^n$ is an $(n-1)$ -rectifiable set and $f \in BV(\mathbb{R}^n)$, then for $\mathcal{H}^{n-1}$ -a.e. point $x \in R$ with normal $\nu_R(x)$, there exist values $f^+(x), f^-(x) \in \mathbb{R}$ such that*

$$\lim_{\rho \rightarrow 0} \frac{1}{\rho^n} \int_{B_\rho^\pm(x)} |f(y) - f^\pm(x)| \, dy = 0,$$

where $B_\rho^\pm(x) = \{y \in B_\rho(x) : \pm(y-x) \cdot \nu_R(x) > 0\}$.

Theorem B.0.5. *Let $u \in BV(\mathbb{R}^n; \mathbb{R}^m)$ and let $f \in C^1(\mathbb{R}^m)$ be a Lipschitz function such that $f(0) = 0$. Then $v = f(u) \in BV(\mathbb{R}^n)$ and its distributional derivative is given by*

$$D(f(u)) = \nabla f(u) \nabla u \mathcal{L}^n + \nabla f(\tilde{u}) D^c u + (f(u^+) - f(u^-)) \nu_u \mathcal{H}^{n-1} \llcorner S_u,$$

where $\tilde{u}$ is the approximate limit of u , and S_u is its jump set.

Applying the chain rule to a product function yields the following decomposition.

Corollary B.0.1. *If $u, v \in BV(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$, then $uv \in BV(\mathbb{R}^n)$ and $D(uv)$ can be decomposed as follows:*

$$\begin{aligned} \nabla(uv) &= u \nabla v + v \nabla u, \\ D^c(uv) &= \tilde{u} D^c v + \tilde{v} D^c u, \\ D^j(uv) &= (u^+ v^+ - u^- v^-) \nu_{S_u \cup S_v} \mathcal{H}^{n-1} \llcorner (S_u \cup S_v). \end{aligned}$$

A particular consequence is the generalized Gauss-Green formula

$$\operatorname{Div} f(E^{(1)}) = \int_{\partial^* E} \operatorname{tr}_E(f) \cdot \nu_E \, d\mathcal{H}^{n-1}$$

for every set of finite perimeter E and $f \in BV(\mathbb{R}^n; \mathbb{R}^n)$, where $\operatorname{Div} f$ denotes the divergence measure of f .

Finally, we include the behavior of sets of finite perimeter under slicing due to Barchiesi, Cagnetti, and Fusco (see [6]).

Theorem B.0.6. *Let $E \subset \mathbb{R}^n$ be a set of finite perimeter. For $(x', y) \in \mathbb{R}^{n-k} \times \mathbb{R}^k$, define $E_{x'} = \{y \in \mathbb{R}^k : (x', y) \in E\}$ and $v_E(x') = \mathcal{H}^k(E_{x'})$. Then $v_E \in BV(\mathbb{R}^{n-k})$ and*

$$P(E) \geq \int_{\mathbb{R}^{n-k}} \sqrt{P(E_{x'})^2 + |\nabla v_E(x')|^2} \, dx' + |D^s v_E|(\mathbb{R}^{n-k}), \quad (\text{B.1})$$

and equality holds whenever E is axially symmetric.

If in addition

$$\mathcal{H}^{n-1}(\{x \in \partial^* E : \nu_E(x) \in \mathbb{R}^{n-k} \times \{0\}\}) = 0,$$

then $v_E \in W^{1,1}(\mathbb{R}^{n-k})$, meaning that $|D^s v_E|(\mathbb{R}^{n-k}) = 0$.

Corollary B.0.2. *Let E be a set of finite perimeter and E^* be its Steiner symmetrization with respect to $\mathbb{R}^{n-k}$. Then $P(E) \geq P(E^*)$.*

Proof. By definition of the Steiner symmetrization, $E_{x'}^*$ is a k -dimensional ball with the same volume as $E_{x'}$, which means that $v_E = v_{E^*}$. The thesis follows from applying the sharp isoperimetric inequality to the slices, which gives $P(E_{x'}) \geq P(E_{x'}^*)$ for $\mathcal{H}^{n-k}$ -a.e. x' . $\square$

Bibliography

- [1] H. Alzer. A new refinement of the Arithmetic Mean-Geometric Mean inequality. *Rocky Mountain Journal of Mathematics*, 27(3):663–667, 1997.
- [2] L. Ambrosio, E. Brué, and D. Semola. *Lectures on Optimal Transport*. UNITEXT. Springer Nature Switzerland, 2024.
- [3] L. Ambrosio, N. Fusco, and D. Pallara. *Functions of Bounded Variation and Free Discontinuity Problems*. Oxford Science Publications. Clarendon Press, 2000.
- [4] T. Aubin. Problèmes isopérimétriques et espaces de Sobolev. *Journal of Differential Geometry*, 11(4):573–598, 1976.
- [5] M. Barchiesi, A. Brancolini, and V. Julin. Sharp dimension free quantitative estimates for the Gaussian isoperimetric inequality. *The Annals of Probability*, 45(2):668–697, 2017.
- [6] M. Barchiesi, F. Cagnetti, and N. Fusco. Stability of the Steiner symmetrization of convex sets. *Journal of the European Mathematical Society*, 15(4):1245–1278, 2013.
- [7] G. Bianchi and H. Egnell. A note on the sobolev inequality. *Journal of Functional Analysis*, 100(1):18–24, 1991.
- [8] V. Bögelein, F. Duzaar, and N. Fusco. A sharp quantitative isoperimetric inequality in higher codimension. *Atti della Accademia Nazionale dei Lincei. Rendiconti Lincei. Matematica e Applicazioni*, 26(3):309–362, 2015.
- [9] V. Bögelein, F. Duzaar, and N. Fusco. A quantitative isoperimetric inequality on the sphere. *Advances in Calculus of Variations*, 10(3):223–265, 2017.
- [10] C. Borell. The Brunn-Minkowski inequality in Gauss space. *Inventiones Mathematicae*, 30(2):207–216, 1975.
- [11] H. J. Brascamp, E. H. Lieb, and J. M. Luttinger. A general rearrangement inequality for multiple integrals. *Journal of Functional Analysis*, 17:227–237, 1974.

- [12] F. Brock. Continuous Steiner symmetrization. *Mathematische Nachrichten*, 172:25–48, 1995.
- [13] F. Brock. Continuous rearrangement and symmetry of solutions of elliptic problems. *Proceedings of the Indian Academy of Sciences - Mathematical Sciences*, 110(2):157–204, 2000.
- [14] S. Campi. Isoperimetric deficit and convex plane sets of maximum translative discrepancy. *Geometriae Dedicata*, 43(1):71–81, 1992.
- [15] E. A. Carlen and C. Kerce. On the cases of equality in Bobkov’s inequality and Gaussian isoperimetric inequalities. *Calculus of Variations and Partial Differential Equations*, 13(1):1–18, 2001.
- [16] E. A. Carlen and M. Loss. Extremals of functionals with competing symmetries. *Journal of Functional Analysis*, 88(2):437–456, 1990.
- [17] O. Chodosh, M. Engelstein, and L. Spolaor. The Riemannian quantitative isoperimetric inequality. *Journal of the European Mathematical Society*, 25(5):1711–1741, 2023.
- [18] M. Cicalese and G. P. Leonardi. A selection principle for the sharp quantitative isoperimetric inequality. *Archive for Rational Mechanics and Analysis*, 206(2):617–643, 2012.
- [19] M. Cicalese and G. P. Leonardi. Best constants for the isoperimetric inequality in quantitative form. *Journal of the European Mathematical Society*, 15(3):1101–1129, 2013.
- [20] F. Dai and Y. Xu. *Approximation Theory and Harmonic Analysis on Spheres and Balls*. Springer Monographs in Mathematics. Springer New York, 2013.
- [21] G. David and S. Semmes. Surfaces quasiminimales de codimension 1 : un morceau de démonstration. *Journées équations aux dérivées partielles*, 1996:1–18, 1996.
- [22] E. De Giorgi. Sulla proprietà isoperimetrica dell’ipersfera, nella classe degli insiemi aventi frontiera orientata di misura finita. *Atti della Accademia Nazionale dei Lincei. Memorie. Classe di Scienze Fisiche, Matematiche e Naturali. Serie 8*, 5(2):33–44, 1958.
- [23] J. Dolbeault, M. J. Esteban, A. Figalli, R. L. Frank, and M. Loss. Sharp stability for Sobolev and log-Sobolev inequalities. *Pure and Applied Analysis*, 5(3):687–734, 2023.
- [24] J. Duoandikoetxea. Reverse Hölder inequalities for spherical harmonics. *Proceedings of the American Mathematical Society*, 101(3):487–491, 1987.

- [25] P. M. N. Feehan. Resolution of singularities and geometric proofs of the Łojasiewicz inequalities. *Geometry & Topology*, 23(7):3273–3313, 2019.
- [26] A. Figalli, N. Fusco, F. Maggi, V. Millot, and M. Morini. Isoperimetry and stability properties of balls with respect to nonlocal energies. *Communications in Mathematical Physics*, 336(1):441–507, 2015.
- [27] A. Figalli, F. Maggi, and A. Pratelli. A mass transportation approach to quantitative isoperimetric inequalities. *Inventiones mathematicae*, 182(1):167–211, 2010.
- [28] B. Fuglede. Stability in the isoperimetric problem for convex or nearly spherical domains in $\mathbb{R}^n$. *Transactions of the American Mathematical Society*, 314(2):619–638, 1989.
- [29] N. Fusco. The quantitative isoperimetric inequality and related topics. *Bulletin of Mathematical Sciences*, 5(3):517–607, 2015.
- [30] N. Fusco and V. Julin. A strong form of the quantitative isoperimetric inequality. *Calculus of Variations and Partial Differential Equations*, 50(3):925–937, 2014.
- [31] N. Fusco, F. Maggi, and A. Pratelli. The sharp quantitative isoperimetric inequality. *Annals of Mathematics*, 168(3):941–980, 2008.
- [32] R. R. Hall. A quantitative isoperimetric inequality in n -dimensional space. *Journal für die reine und angewandte Mathematik*, 428:161–176, 1992.
- [33] R. R. Hall and W. K. Hayman. A problem in the theory of subordination. *Journal d’Analyse Mathématique*, 60:99–111, 1993.
- [34] B. Kawohl. *Rearrangements and Convexity of Level Sets in PDE*, volume 1150 of *Lecture Notes in Mathematics*. Springer Berlin, Heidelberg, 1985.
- [35] D. A. La Manna. Local minimality of the ball for the Gaussian perimeter. *Advances in Calculus of Variations*, 12:193–210, 2019.
- [36] E. H. Lieb and M. Loss. *Analysis*, volume 14 of *Graduate Studies in Mathematics*. American Mathematical Society, 2001.
- [37] F. Maggi. Some methods for studying stability in isoperimetric type problems. *Bulletin of the American Mathematical Society*, 45(3):367–408, 2008.
- [38] A. Segal. Remark on stability of Brunn–Minkowski and isoperimetric inequalities for convex bodies. In B. Klartag, S. Mendelson, and V. D. Milman, editors, *Geometric Aspects of Functional Analysis: Israel Seminar 2006–2010*, volume 2050 of *Lecture Notes in Mathematics*, pages 381–391. Springer, Berlin, Heidelberg, 2012.

- [39] V. N. Sudakov and B. S. Tsirelson. Extremal properties of half-spaces for spherically invariant measures. *Zapiski Nauchnykh Seminarov LOMI*, 41:14–24, 1974. English translation in *Journal of Soviet Mathematics*, 9(1):9–18, 1978.
- [40] G. Talenti. Best constant in Sobolev inequality. *Annali di Matematica Pura ed Applicata*, 110(1):353–372, 1976.
- [41] I. Tamanini. Regularity results for almost minimal oriented hypersurfaces in $\mathbb{R}^n$. *Quaderni del Dipartimento di Matematica dell'Università del Salento*, 1:1–92, 1984.